6 Water Quality Impact

6.1 Introduction

6.1.1.1 This section presents the assessment of potential water quality impacts, which may arise during the construction and operation of the Project. Mitigation measures will be implemented as far as practicable to control the potential water quality impact during the construction and operational phase.

6.1.1.2 The water quality impact assessment has been conducted in accordance with the requirements of Annexes 6 and 14 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) and EIA Study Brief (ref: EIA SB No: ESB-246/2012 dated 18 June 2012).

6.2 Environmental Legislation, Standards and Guidelines

6.2.1.1 The relevant legislations, standards and guidelines applicable to present study for the assessment of water quality impacts include:

- Water Pollution Control Ordinance (WPCO) (Cap. 358);
- Technical Memorandum for Effluents Discharged into Drainage and Sewerage Systems Inland and Coastal Waters (TM-DSS)
- EIAO-TM;
- No Net Increase in Pollution Loads Requirement in Deep Bay;
- Hong Kong Planning Standards and Guidelines (HKPSG); and
- Practice Note for Professional Persons on Construction Site Drainage (ProPECC PN 1/94)

Water Pollution Control Ordinance (WPCO) (Cap. 358)

6.2.1.2 The entire Hong Kong waters are divided into ten Water Control Zones (WCZs) and four supplementary WCZs under the Water Pollution Control Ordinance (WPCO) (Cap. 358). Each WCZ has a designated set of statutory Water Quality Objectives (WQOs) designed to protect the inland and/or marine environment and its users. The Project is located in the Deep Bay WCZ and may have potential impacts to the North Western WCZ. The corresponding WQOs are summarised in the table below.

Table 6.1 Water Quality Objectives for Deep Bay Water Control Zones

Parameters	Objectives	Sub-Zone
Aesthetic appearance	(a) Waste discharges shall cause no objectionable odours or discolouration of the water.	Whole zone
	(b) Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent.	Whole zone
	(c) Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam.	Whole zone
	(d) There should be no recognisable sewage-derived debris.	Whole zone
	(e) Floating, submerged and semi-submerged	Whole zone

Parameters	Objectives	Sub-Zone
	objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent.	
	(f) Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits.	Whole zone
Bacteria	(a) The level of *Escherichia coli* should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year.	Secondary Contact Recreation Subzone and Mariculture Subzone (L.N. 455 of 1991)
	(b) The level of *Escherichia coli* should be zero per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(c) The level of *Escherichia coli* should not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
	(d) The level of *Escherichia coli* should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive in one calendar year. Samples should be taken at least 3 times in a calendar month at intervals of between 3 and 14 days.	Yung Long Bathing Beach Subzone (L.N. 455 of 1991)
Colour	(a) Waste discharges shall not cause the colour of water to exceed 30 Hazen units.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the colour of water to exceed 50 Hazen units.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Dissolved Oxygen	(a) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 milligrams per litre for 90% of the sampling occasions during the year; values should be taken at 1 metre below surface.	Inner Marine Subzone excepting Mariculture Subzone
	(b) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 milligrams per litre for 90% of the sampling occasions during the year; values should be calculated as water column average (arithmetic mean of at least 2 measurements at 1 metre below surface and 1 metre above seabed). In addition, the concentration of dissolved oxygen should not be less than 2 milligrams per litre within 2 metres of the seabed for 90% of the sampling occasions during the year.	Outer Marine Subzone excepting Mariculture Subzone
	(c) The dissolved oxygen level should not be less than 5 milligrams per litre for 90% of the sampling	Mariculture Subzone

Parameters	Objectives	Sub-Zone
	occasions during the year; values should be taken at 1 metre below surface.	
	(d) Waste discharges shall not cause the level of dissolved oxygen to be less than 4 milligrams per litre.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone, Water Gathering Ground Subzones and other inland waters of the Zone
pH	(a) The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units.	Marine waters excepting Yung Long Bathing Beach Subzone
	(b) Waste discharges shall not cause the pH of the water to exceed the range of 6.5-8.5 units.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(c) The pH of the water should be within the range of 6.0-9.0 units.	Other inland waters
	(d) The pH of the water should be within the range of 6.0-9.0 units for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units.	Yung Long Bathing Beach Subzone
Temperature	Waste discharges shall not cause the natural daily temperature range to change by more than 2.0 degrees Celsius.	Whole Zone
Salinity	Waste discharges shall not cause the natural ambient salinity level to change by more than 10%	Whole Zone
Suspended solids	(a) Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of suspended solids which may adversely affect aquatic communities.	Marine waters
	(b) Waste discharges shall not cause the annual median of suspended solids to exceed 20 milligrams per litre.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Ganges Subzone, Indus Subzone, Water Gathering Ground Subzones and other inland waters
Ammonia	The un-ionized ammoniacal nitrogen level should not be more than 0.021 milligram per litre, calculated as the annual average (arithmetic mean).	Whole Zone
Nutrients	(a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants.	Inner and Outer Marine Subzones
	(b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.7 milligram per litre, expressed as annual mean.	Inner Marine Subzone

Parameters	Objectives	Sub-Zone
	(c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 milligram per litre, expressed as annual water column average (arithmetic mean of at least 2 measurements at 1 metre below surface and 1 metre above seabed).	Outer Marine Subzone
5 day biochemical oxygen demand	(a) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 milligrams per litre.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 milligrams per litre.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Chemical oxygen demand	(a) Waste discharges shall not cause the chemical oxygen demand to exceed 15 milligrams per litre.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the chemical oxygen demand to exceed 30 milligrams per litre.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Toxins	(a) Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other.	Whole Zone
	(b) Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment.	Whole Zone
Phenol	Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 milligrams per litre as C6H5OH.	Yung Long Bathing Beach Subzone
Turbidity	Waste discharges shall not reduce light transmission substantially from the normal level.	Yung Long Bathing Beach Subzone

Table 6.2 Water Quality Objectives for North Western Water Control Zones

Parameters	Objectives	Sub-Zone
Aesthetic appearance	(a) Waste discharges shall cause no objectionable odours or discolouration of the water.	Whole Zone
	(b) Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent.	Whole Zone
	(c) Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam.	Whole Zone
	(d) There should be no recognisable sewage-derived debris.	Whole Zone

Parameters	Objectives	Sub-Zone
	(e) Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent.	Whole Zone
	(f) Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits.	Whole Zone
Bacteria	(a) The level of Escherichia coli should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year.	Secondary Contact Recreation Subzones
	(b) The level of Escherichia coli should be less than 1 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Tuen Mun (A) and Tuen Mun (B) Subzones and Water Gathering Ground Subzones
	(c) The level of Escherichia coli should not exceed 1000 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Tuen Mun (C) Subzone and other inland waters
	(d) The level of Escherichia coli should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive. Samples should be taken at least 3 times in one calendar month at intervals of between 3 and 14 days.	Bathing Beach Subzones
Colour	(a) Waste discharges shall not cause the colour of water to exceed 30 Hazen units.	Tuen Mun (A) and Tuen Mun (B) Subzones and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the colour of water to exceed 50 Hazen units.	Tuen Mun (C) Subzone and other inland waters
Dissolved oxygen	(a) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 mg per litre for 90% of the sampling occasions during the whole year; values should be calculated as water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed). In addition, the concentration of dissolved oxygen should not be less than 2 mg per litre within 2 m of the seabed for 90% of the sampling occasions during the whole year	Marine waters
	(b) Waste discharges shall not cause the level of dissolved oxygen to be less than 4 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones, Water Gathering Ground Subzones and other inland waters
pH	(a) The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units.	Marine waters excepting Bathing Beach Subzones

Parameters	Objectives	Sub-Zone
	(b) Waste discharges shall not cause the pH of the water to exceed the range of 6.5-8.5 units.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(c) The pH of the water should be within the range of 6.0-9.0 units.	Other inland waters
	(d) The pH of the water should be within the range of 6.0-9.0 units for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units.	Bathing Beach Subzones
Temperature	Waste discharges shall not cause the natural daily temperature range to change by more than 2.0 degrees Celsius.	Whole Zone
Salinity	Waste discharges shall not cause the natural ambient salinity level to change by more than 10%	Whole Zone
Suspended solids	(a) Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of suspended solids which may adversely affect aquatic communities.	Marine waters'
	(b) Waste discharges shall not cause the annual median of suspended solids to exceed 20 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(c) Waste discharges shall not cause the annual median of suspended solids to exceed 25 mg per litre.	Other inland waters
Ammonia	The un-ionised ammoniacal nitrogen level should not be more than 0.021 milligram per litre, calculated as the annual average (arithmetic mean).	Whole Zone
Nutrients	(a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants.	Marine waters
	(b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.3 mg per litre, expressed as annual water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed).	Castle Peak Bay Subzone
	(c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 mg per litre, expressed as annual water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed).	Marine waters excepting Castle Peak Bay Subzone
5-day biochemical oxygen demand	(a) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones

Parameters	Objectives	Sub-Zone
	(b) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 milligrams per litre.	Other inland waters
Chemical oxygen demand	(a) Waste discharges shall not cause the chemical oxygen demand to exceed 15 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the chemical oxygen demand to exceed 30 milligrams per litre.	Other inland waters
Toxins	(a) Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic, carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other.	Whole Zone
	(b) Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment.	Whole Zone
Phenol	Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 milligrams per litre as C6H5OH.	Bathing Beach Subzones
Turbidity	Waste discharges shall not reduce light transmission substantially from the normal level.	Bathing Beach Subzones

Technical Memorandum for Effluents Discharge into Drainage and Sewerage Systems, Inland & Coastal Waters

6.2.1.3 Apart from the WQOs, Section 21 of the WPCO also specifies the limits to control the physical, chemical and microbial parameters for effluent discharges into drainage and sewage system at both inland and coastal waters under the TM-DSS. The discharge limits vary with the effluent flow rates and the sewage from the Project treated after sewage treatment works (STW) should comply with the standards for effluent discharged into inland waters, which would be used for either irrigation, pond fish culture or amenity subject to the exact locations. Group B (for irrigation), C (for pond fish culture) and D (for amenity) inland water standards in TM-DSS are therefore adopted and the effluent discharge standards are presented in tables below.

Table 6.3 Standards for effluents discharged into Group B Inland Waters

Parameter	Flow rate (m^3/day)							
	≤ 200	**> 200 & ≤400**	**> 400 & ≤ 600**	**> 600 & ≤ 800**	**> 800 & ≤ 1000**	**> 1000 & ≤ 1500**	**> 1500 & ≤ 2000**	**> 2000 & ≤ 3000**
pH (pH units)	6.5-8.5	6.5-8.5	6.5-8.5	6.5-8.5	6.5-8.5	6.5-8.5	6.5-8.5	6.5-8.5
Temperature (℃)	35	30	30	30	30	30	30	30

Parameter	Flow rate (m^3/day)							
	≤ 200	> 200 & ≤400	> 400 & ≤ 600	> 600 & ≤ 800	> 800 & ≤ 1000	> 1000 & ≤ 1500	> 1500 & ≤ 2000	> 2000 & ≤ 3000
Colour (lovibond units) (25mm cell length)	1	1	1	1	1	1	1	1
Suspended solids	30	30	30	30	30	30	30	30
BOD	20	20	20	20	20	20	20	20
COD	80	80	80	80	80	80	80	80
Oil & Grease	10	10	10	10	10	10	10	10
Iron	10	8	7	5	4	3	2	1
Boron	5	4	3	2.5	2	1.5	1	0.5
Barium	5	4	3	2.5	2	1.5	1	0.5
Mercury	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Cadmium	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Selenium	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1
Other toxic metals individually	0.5	0.5	0.2	0.2	0.2	0.1	0.1	0.1
Total Toxic metals	2	1.5	1	0.5	0.5	0.2	0.2	0.2
Cyanide	0.1	0.1	0.1	0.08	0.08	0.05	0.05	0.03
Phenols	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Sulphide	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Fluoride	10	10	8	8	8	5	5	3
Sulphate	800	800	600	600	600	400	400	400
Chloride	1000	1000	800	800	800	600	600	400
Total phosphorus	10	10	10	8	8	8	5	5
Ammonia nitrogen	5	5	5	5	5	5	5	5
Nitrate + nitrite nitrogen	30	30	30	20	20	20	10	10
Surfactants (total)	5	5	5	5	5	5	5	5
E. coli (count/100ml)	100	100	100	100	100	100	100	100

Notes:
[1] All units in mg/L unless otherwise stated

Table 6.4 Standards for effluents discharged into Group C Inland Waters

Parameter	Flow rate (m^3/day)			
	≤ 100	**> 100 & ≤500**	**> 500 & ≤ 1000**	**> 1000 & ≤ 2000**
pH (pH units)	6-9	6-9	6-9	6-9
Temperature (°C)	30	30	30	30
Colour (lovibond units) (25mm cell length)	1	1	1	1
Suspended solids	20	10	10	5
BOD	20	15	10	5
COD	80	60	40	20
Oil & Grease	1	1	1	1
Boron	10	5	4	2
Barium	1	1	1	0.5
Iron	0.5	0.4	0.3	0.2
Mercury	0.001	0.001	0.001	0.001
Cadmium	0.001	0.001	0.001	0.001
Silver	0.1	0.1	0.1	0.1
Copper	0.1	0.1	0.05	0.05
Selenium	0.1	0.1	0.05	0.05
Lead	0.2	0.2	0.2	0.1
Nickel	0.2	0.2	0.2	0.1
Other toxic metals individually	0.5	0.4	0.3	0.2
Total Toxic metals	0.5	0.4	0.3	0.2
Cyanide	0.05	0.05	0.05	0.01
Phenols	0.1	0.1	0.1	0.1
Sulphide	0.2	0.2	0.2	0.1
Fluoride	10	7	5	4
Sulphate	800	600	400	200
Chloride	1000	1000	1000	1000
Total phosphorus	10	10	8	8
Ammonia nitrogen	2	2	2	1
Nitrate + nitrite nitrogen	30	30	20	20
Surfactants (total)	2	2	2	1
E. coli (count/100ml)	1000	1000	1000	1000

Notes:
[1] All units in mg/L unless otherwise stated

Table 6.5 Standards for effluents discharged into Group D Inland Waters

Parameter	Flow rate (m^3/day)							
	≦200	**＞200 and ≦400**	**＞400 and ≦600**	**＞600 and ≦800**	**＞800 and ≦1000**	**＞ 1000 and ≦1500**	**＞ 1500 and ≦2000**	**＞ 2000 and ≦3000**
pH (pH units)	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10
Temperature (℃)	30	30	30	30	30	30	30	30
Colour (lovibond units) (25mm cell length)	1	1	1	1	1	1	1	1
Suspended solids	30	30	30	30	30	30	30	30
BOD	20	20	20	20	20	20	20	20
COD	80	80	80	80	80	80	80	80
Oil & Grease	10	10	10	10	10	10	10	10
Iron	10	8	7	5	4	2.7	2	1.3
Boron	5	4	3.5	2.5	2	1.5	1	0.7
Barium	5	4	3.5	2.5	2	1.5	1	0.7
Mercury	0.1	0.05	0.001	0.001	0.001	0.001	0.001	0.001
Cadmium	0.1	0.05	0.001	0.001	0.001	0.001	0.001	0.001
Other toxic metals individually	1	1	0.8	0.8	0.5	0.5	0.2	0.2
Total Toxic metals	2	2	1.6	1.6	1	1	0.5	0.4
Cyanide	0.4	0.4	0.3	0.3	0.2	0.1	0.1	0.05
Phenols	0.4	0.3	0.2	0.1	0.1	0.1	0.1	0.1
Sulphide	1	1	1	1	1	1	1	1
Sulphate	800	600	600	600	600	400	400	400
Chloride	1000	800	800	800	600	600	400	400
Fluoride	10	8	8	8	5	5	3	3
Total phosphorus	10	10	10	8	8	8	5	5
Ammonia nitrogen	20	20	20	20	20	20	20	10
Nitrate + nitrite nitrogen	50	50	50	30	30	30	30	20
Surfactants (total)	15	15	15	15	15	15	15	15
E. coli (count/100ml)	1000	1000	1000	1000	1000	1000	1000	1000

Notes:
[1] All units in mg/L unless otherwise stated

6.2.1.4 The TM-DSS also specifies the discharge standards into foul sewers leading into the Government's sewage treatment plants as below tables. Subject to the flow rate of the effluents, corresponding standards for the effluent discharge into the Government's foul sewers should be followed.

Table 6.6 Standards for effluents discharged into foul sewers leading into the Government's sewage treatments plants

Parameter	Flow rate (m^3/day)												
	≤ 10	**> 10 & ≤100**	**> 100 & ≤ 200**	**> 200 & ≤ 400**	**> 400 & ≤ 600**	**> 600 & ≤ 800**	**> 800 & ≤ 1000**	**> 1000 & ≤ 1500**	**> 1500 & ≤ 2000**	**> 2000 & ≤ 3000**	**> 3000 & ≤ 4000**	**> 4000 & ≤ 5000**	**> 5000 & ≤ 6000**
pH (pH units)	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10	6-10
Temperature (℃)	43	43	43	43	43	43	43	43	43	43	43	43	43
Suspended solids	1200	1000	900	800	800	800	800	800	800	800	800	800	800
Settleable solids	100	100	100	100	100	100	100	100	100	100	100	100	100
BOD	1200	1000	900	800	800	800	800	800	800	800	800	800	800
COD	3000	2500	2200	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000
Oil & Grease	100	100	50	50	50	40	30	20	20	20	20	20	20
Iron	30	25	25	25	15	12.5	10	7.5	5	3.5	2.5	2	1.5
Boron	8	7	6	5	4	3	2.4	1.6	1.2	0.8	0.6	0.5	0.4
Barium	8	7	6	5	4	3	2.4	1.6	1.2	0.8	0.6	0.5	0.4
Mercury	0.2	0.15	0.1	0.1	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Cadmium	0.2	0.15	0.1	0.1	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Copper	4	4	4	3	1.5	1.5	1	1	1	1	1	1	1
Nickel	4	3	3	2	1.5	1.5	1	0.8	0.7	0.7	0.6	0.6	0.6
Chromium	2	2	2	2	1	0.7	0.6	0.4	0.3	0.2	0.1	0.1	0.1
Zinc	5	5	4	3	1.5	1.5	1	0.8	0.7	0.7	0.6	0.6	0.6
Silver	4	3	3	2	1.5	1.5	1	0.8	0.7	0.7	0.6	0.6	0.6
Other toxic metals individually	2.5	2.2	2	1.5	1	0.7	0.6	0.4	0.3	0.2	0.15	0.12	0.1
Total toxic metals	10	10	8	7	3	2	2	1.6	1.4	1.2	1.2	1.2	1
Cyanide	2	2	2	1	0.7	0.5	0.4	0.27	0.2	0.13	0.1	0.08	0.06
Phenols	1	1	1	1	0.7	0.5	0.4	0.27	0.2	0.13	0.1	0.1	0.1
Sulphide	10	10	10	10	5	5	4	2	2	2	1	1	1
Sulphate	1000	1000	1000	1000	1000	1000	1000	900	800	600	600	600	600

Parameter	Flow rate (m^3/day)												
	≤ 10	**> 10 & ≤100**	**> 100 & ≤ 200**	**> 200 & ≤ 400**	**> 400 & ≤ 600**	**> 600 & ≤ 800**	**> 800 & ≤ 1000**	**> 1000 & ≤ 1500**	**> 1500 & ≤ 2000**	**> 2000 & ≤ 3000**	**> 3000 & ≤ 4000**	**> 4000 & ≤ 5000**	**> 5000 & ≤ 6000**
Total nitrogen	200	200	200	200	200	200	200	100	100	100	100	100	100
Total phosphorus	50	50	50	50	50	50	50	25	25	25	25	25	25
Surfactants (total)	200	150	50	40	30	25	25	25	25	25	25	25	25

Notes:
[1] All units in mg/L unless otherwise stated

Table 6.7 Standards for effluents discharged into foul sewers leading into the Government's sewage treatments plants with microbial treatment

Parameter	Flow rate (m^3/day)												
	≤ 10	**> 10 & ≤100**	**> 100 & ≤ 200**	**> 200 & ≤ 400**	**> 400 & ≤ 600**	**> 600 & ≤ 800**	**> 800 & ≤ 1000**	**> 1000 & ≤ 1500**	**> 1500 & ≤ 2000**	**> 2000 & ≤ 3000**	**> 3000 & ≤ 4000**	**> 4000 & ≤ 5000**	**> 5000 & ≤ 6000**
Copper	1.5	1	1	1	0.8	0.6	0.5	0.4	0.3	0.2	0.15	0.1	0.05

Notes:
[1] All units in mg/L unless otherwise stated

Environmental Impact Assessment Ordinance (Cap. 499), Technical Memorandum on Environmental Impact Assessment Process

6.2.1.5 The EIAO-TM specifies the assessment methods and criteria for impact assessment. This Study follows the EIAO-TM to assess the potential water quality impact that may arise during both the construction and operational phases of the Project. Sections in the EIAO-TM relevant to the water quality impact assessment are:

- Annex 6 - Criteria for Evaluating Water Pollution; and
- Annex 14 - Guidelines for Assessment of Water Pollution

No Net Increase in Pollution Loads Requirement in Deep Bay

6.2.1.6 The 'No Net Increase in Pollution Loads Requirement' aims to provide protection to the inland and marine water quality of the Deep Bay WCZ. According to EPD's "Deep Bay Water Quality Regional Control Strategy Study", the pollution load entering into Deep Bay have exceeded the assimilative capacity of the water body. Further increasing the pollution loads to the water body is therefore environmentally undesirable.

6.2.1.7 According to Item (xiv) of Appendix D of the EIA Study Brief, no net increase in pollution load to Deep Bay will be required for the Project.

Hong Kong Planning Standards and Guidelines

6.2.1.8 Chapter 9 of the HKPSG outlines environmental requirements that need to be considered in land use planning. The recommended guidelines, standards and guidance cover the selection of suitable locations for the developments and sensitive uses, provision of environmental facilities, and design, layout, phasing and operational controls to minimise adverse environmental impacts. It also lists out environmental factors that influence land use planning and recommends buffer distances for land uses.

ProPECC PN 1/94 "Construction Site Drainage"

6.2.1.9 The ProPECC PN1/94 provides guidelines for the handling and disposal of construction discharges. It is applicable to this study for the control of site runoff and wastewater generated during the construction phase. The types of discharges from construction sites outlined in the ProPECC PN1/94 include:

- Surface runoff;
- Groundwater;
- Boring and drilling water;
- Wastewater from concrete batching plant;
- Wheel washing water;
- Bentonite slurries;
- Water for testing and sterilization of water retaining structures and water pipes;
- Wastewater from building construction and site facilities; and
- Acid cleaning, etching and pickling wastewater.

6.3 Sensitive Receivers

6.3.1.1 In order to identify the water sensitive receivers (WSRs), a desktop study including review of Tong Yan San Tsuen (TYST) OZP (S/YL-TYST/10), Tai Tong OZP (S/YL-TT/16), Yuen Long OZP (S/YL/23), topographic maps and aerial photos has been conducted together with site visits. The WSRs in the vicinity of the Project are presented in **Table 6.8** and illustrated in **Figure 6.1**, **Figure 6.1a**, **Figure 6.1b, Figure 6.1c** and **Figure 6.1d**.

Table 6.8 Water Sensitive Receivers

ID	Description	Within PDA/ Project Area (Yes/ No)	Type	Status	Remarks
Streams					
S01	Yuen Long Nullah	Yes	Watercourse	• Channelised	Retained
S02	Near TYST	Yes	Watercourse	• Largely natural • Fragmented	Retained and rezoned as GB for conservation The lower section of S02 will be diverted to S03
S03	Near Windsor Garden	Yes	Watercourse	• Largely natural • Fragmented	Retained and rezoned as GB for conservation
S04	Near proposed YLS STW	Yes	Watercourse	• Semi-natural	Retained and rezoned as GB for conservation
S05	Yeung Ka Tsuen Ecologically Important Stream (EIS)	No	EIS	• Semi-natural	-
S06	Near Wong Nai Tun Tsuen	No	Watercourse	• Semi-natural • Fragmented	-
S07	Near One Hyde Park	Yes (partially)	Watercourse	• Heavily modified • Fragmented	The section within the PDA will be permanently lost
S08	Near One Hyde Park	No	Watercourse	• Heavily modified • Fragmented	-
S09	Near Lam Tai West Road	Yes	Watercourse	• Largely natural • Fragmented	Permanently lost
S10	Near Chun Wah Villas Phase 2	No	Watercourse	• Semi-natural • Fragmented	-
S11	At Shap Pat Heung	No	Watercourse	• Semi-natural • Fragmented	-
S12	Near Sheung Yau Tin Tsuen	No	Watercourse	• Semi-natural • Fragmented	-
S13	Near Chuk San Tsuen	No	Watercourse	• Channelised	-
S14	Near Yeung Uk Tsuen	No	Watercourse	• Semi-natural • Fragmented	-
S15	Near Chung Sing Villa	No	Watercourse	• Semi-natural • Fragmented	-

ID	Description	Within PDA/ Project Area (Yes/ No)	Type	Status	Remarks
S16	Near Pok Oi Interchange	No	Watercourse	• Semi-natural • Fragmented	-
S17	Along Kiu Hung Road	Yes (partially)	Watercourse	• Channelised	Retained
S18	Shan Pui River	No	Watercourse	• Channelised	-
S19	Near the southwest of proposed Reclaimed Water Service Reservoir	No	Watercourse	• Channelised	-
S20	Near the north of proposed Reclaimed Water Service Reservoir	Yes (partially)	Watercourse	• Semi-natural • Fragmented	The section within the PDA will be permanently lost
S21	Near One Hyde Park	Yes	Watercourse	• Heavily modified • Fragmented	Permanently lost
S22	Near Chun Wah Villas Phase II	No	Watercourse	• Semi-natural • Fragmented	-
S23	Near the southern end of Lam Tai East Road	Yes (partially)	Watercourse	• Largely natural • Fragmented	The section within the PDA will be permanently lost
S24	Near the southern end of Lam Tai East Road	No	Watercourse	• Largely natural • Fragmented	-
S25	Near the southern end of Lam Tai East Road	Yes (partially)	Watercourse	• Largely natural • Fragmented	The section within the PDA will be permanently lost
S26	Near One Hyde Park	Yes	Watercourse	• Heavily modified • Fragmented	Permanently lost
S27	Near One Hyde Park	Yes	Watercourse	• Heavily modified • Fragmented	Permanently lost
S28	Immediate upstream of the southern end of Yuen Long Nullah	Yes (partially)	Watercourse	• Largely natural • Fragmented	The section within the PDA will be permanently lost
S29	Along Hung Tin Road	No	Watercourse	• Semi-natural	-
S30	Along Shek Po Road	No	Watercourse	• Channelised	-
S31	South of San Wai STW	No	Watercourse	• Semi-natural	-
S32	Near Sik Kong Wai	No	Watercourse	• Semi-natural	-

ID	Description	Within PDA/ Project Area (Yes/ No)	Type	Status	Remarks
S33	Near Sik Kong Wai	No	Watercourse	• Semi-natural	-
S34	North of San Wai STW	No	Watercourse	• Semi-natural	-
Marine					
M01	Inner Deep Bay	No	Marine waters	• Open Sea	-
Ponds					
P01	Near Yeung Ka Tsuen	No	Pond	• Appeared to be a former farm pond for irrigation	-
P02	Near Fui Sha Wai	No	Pond	• Fung Shui Pond/ Landscaped Pond	-
P03	~250m west of southern end of Long Hon Road	No	Pond	• Possibly active for aquaculture	-
P04	~250m west of southern end of Long Hon Road	No	Pond	• Possibly active for aquaculture	-
P05	Near Sha Tseng Tsuen	Yes	Pond	• Active farm pond for irrigation only	Permanently lost
P06	Near Sha Tseng Tsuen	Yes	Pond	• Active farm pond for irrigation only	Permanently lost
P07	Near Sha Tseng Tsuen	Yes	Pond	• Active farm pond for irrigation only	Permanently lost
P08	Near Sha Tseng Tsuen	Yes	Pond	• Active farm pond for irrigation only	Permanently lost
P09	Near Sha Tseng Tsuen	Yes	Pond	• Active farm pond for irrigation only	Permanently lost
P10	South of Shan Ha Tsuen	No	Pond	• Appeared to be a former farm pond for irrigation	-
P11	Within woodland adjacent to Tan Kwai Tsuen Fresh Water Service Reservoir	No	Pond	• Abandoned fish pond [1]	-
P12	Near Fui Sha Wai	No	Pond	• Active farm pond for irrigation only	-
P13	Near Yeung Uk Tsuen	No	Pond	• Appeared to be a former farm pond for irrigation	-
P17	~80m east of middle of Lam Tai East Road	Yes	Pond	• Managed pond for non-aquaculture use	Permanently lost
P18	Near Wong Nai Tun Tsuen	No	Pond	• Overgrown with vegetation	-

ID	Description	Within PDA/ Project Area (Yes/ No)	Type	Status	Remarks
P19	Near Tai Tong Tsuen	No	Pond	• Overgrown with vegetation	-
P21	Near Ying Lung Wai	No	Pond	• Fung Shui Pond/ Landscaped Pond	-
P22	Near Fui Sha Wai	No	Pond	• Inactive fish pond [2]	-
P23	~300m near the southern end of Kung Um Road	Yes	Pond	• Overgrown with vegetation	Permanently lost
P24	~300m near the southern end of Kung Um Road	Yes	Pond	• Overgrown with vegetation	Permanently lost
P25	Near Kong Tau Tsuen	No	Pond	• Overgrown with vegetation	-
P26	Near Kong Tau Tsuen	No	Pond	• Overgrown with vegetation	-
P27	Near Tai Kei Leng	No	Pond	• Overgrown with vegetation	-
P36	Near Yeung Ka Tsuen	No	Pond	• Overgrown with vegetation	-
P37	Near the southern end of Long Hon Road	Yes	Pond	• Overgrown with vegetation	Permanently lost
P39	At Ma Tin Tsuen	No	Pond	• Stormwater pond	-
P40	~600m northwest of San Sang San Tsuen	No	Pond	• Landscaped Pond	-
P41	~600m northwest of San Sang San Tsuen	No	Pond	• Overgrown with vegetation	-
P42	~600m northwest of San Sang San Tsuen	No	Pond	• Active farm pond for irrigation only	-
P43	~500m north of San Sang San Tsuen under Western Corridor	No	Pond	• Managed pond for non-aquaculture use	-
P44	~500m north of San Sang San Tsuen under Western Corridor	No	Pond	• Managed pond for non-aquaculture use	-
P45	~500m north of San Sang San Tsuen under Western Corridor	No	Pond	• Overgrown with vegetation	-
P46	~500m north of San Sang San Tsuen under Western Corridor	No	Pond	• Managed pond for non-aquaculture use	-

ID	Description	Within PDA/ Project Area (Yes/ No)	Type	Status	Remarks
P47	~500m north of San Sang San Tsuen under Western Corridor	No	Pond	• Managed pond for non-aquaculture use	-
P48	At Sik Kong Wai	No	Pond	• Stormwater pond	-
P49	At Sik Kong Wai	No	Pond	• Stormwater pond	-
P50	At Sik Kong Tsuen	No	Pond	• Stormwater pond	-
P51	At Lo Uk Tsuen	No	Pond	• Stormwater pond	-

Note:
[1] Abandoned fish pond refers to pond in which there is physical evidence that aquaculture has not taken place for many years and/or where there are obvious physical constraints to the resumption of fisheries activities.
[2] Inactive fish pond refers to pond with no current commercial aquaculture activities, but no major physical constraints to the resumption of such activity in the short-term.

Planned WSRs

6.3.1.2 Other than the above existing WSRs, the current RODP has proposed a number of initiatives at certain locations which would eventually become WSRs. These WSRs include the following, and their locations are shown in **Figure 6.1**:

- Diversion of S02 to Yuen Long (West) Nullah
- Reedbed
- Hillside river corridor and retention lake

6.4 Baseline Conditions

6.4.1 General Description

6.4.1.1 The Project Site boundary falls within the Deep Bay WCZ according to the WPCO. It is located at upstream of Yuen Long Creek along Kung Um Road. Downstream of the project is a drainage channel along the edge of Tai Tseng Wai and Yuen Long Creek and the ultimate discharge of this drainage channel is downstream of Shan Pui River and Inner Deep Bay. Apart from Yuen Long Creek, the inner Deep Bay was most affected by the discharges from Shenzhen River as well as Kam Tin River and Tin Shui Wai Nullah. The WQO compliance rate of the Deep Bay WCZ increased from 40% in 2008 to 53% in 2015.

6.4.2 Baseline Water Quality Conditions

EPD's River Monitoring Stations

6.4.2.1 The nearest water quality monitoring stations are Yuen Long Creek (YL1 -YL4) and **Table 6.9** summarises the water quality monitoring data for YL1 -YL4.

Table 6.9 Water Quality at Yuen Long Creek (YL1 -YL4) for Year 2015

Parameter	YL1	YL2	YL3	YL4
Dissolved oxygen (mg/L)	5.5 (4.0 - 7.9)	6 (4.7 - 11.4)	3.9 (2.0 - 8.3)	4.6 (3.0 - 6.7)

Parameter	YL1	YL2	YL3	YL4
pH	7.4 (7.1 - 7.6)	7.3 (7.1 - 7.9)	7.5 (7.3 - 9.7)	7.3 (7.0 - 7.6)
SS (mg/L)	18 (4 - 91)	4 (<1 - 17)	34 (11 - 130)	50 (11 - 98)
5-day Biochemical Oxygen Demand (mg/L)	25 (8 - 41)	7 (4 - 12)	58 (29 - 131)	140 (18 - 210)
Chemical Oxygen Demand (mg/L)	26 (15 - 61)	26 (18 - 35)	48 (17 - 190)	107 (28 - 210)
Oil & grease (mg/L)	0.6 (<0.5 - 3.3)	0.6 (<0.5 - 2.7)	1.4 (<0.5 - 6.6)	6.2 (<0.5 - 18.0)
Faecal coliforms (cfu/100mL)	400,000 (110,000 - 970,000)	81,000 (24,000 - 490,000)	2,400,000 (920,000 - 7,400,000)	5,000,000 (1,300,000 - 12,000,000)
***E. coli*(cfu/100mL)[4]**	180,000 (71,000 - 660,000)	47,000 (9,000 - 340,000)	1,000,000 (260,000 - 2,500,000)	2,000,000 (510,000 - 5,100,000)
Ammonia-nitrogen (mg/L)	7.6 (1.20 - 18.00)	10 (2.00 - 14.00)	6.15 (2.20 - 12.00)	7.4 (1.60 - 8.40)
Nitrate-nitrogen (mg/L)	0.46 (<0.01 - 1.10)	2.45 (1.00 - 9.40)	0.01 (<0.01 - 1.40)	<0.01 (<0.01 - 0.59)
Total Kjeldahl nitrogen (mg/L)	11.5 (3.20 - 23.00)	12.5 (4.40 - 19.00)	10 (4.40 - 23.00)	12 (3.20 - 29.00)
Ortho-phosphate (mg/L)	1.2 (0.46 - 2.50)	2.55 (0.59 - 3.80)	0.92 (0.28 - 3.00)	0.43 (0.06 - 0.75)
Total phosphorus (mg/L)	1.8 (0.77 - 3.00)	2.8 (0.85 - 4.00)	1.35 (0.58 - 3.70)	1.35 (0.59 - 2.00)
Total sulphide (mg/L)	0.03 (<0.02 - 0.13)	<0.02 (<0.02 - 0.04)	0.08 (<0.02 - 0.10)	0.12 (<0.02 - 0.33)
Aluminium (µg/L)	133 (99 - 371)	51 (<50 - 221)	165 (79 - 1,384)	193 (139 - 372)
Cadmium (µg/L)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)
Chromium (µg/L)	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 2)	1 (<1 - 2)
Copper (µg/L)	7 (5 - 11)	3 (2 - 8)	8 (4 - 23)	6 (3 - 9)
Lead (µg/L)	2 (2 - 15)	<1 (<1 - 2)	3 (1 - 17)	3 (1 - 9)
Zinc (µg/L)	53 (39 - 91)	32 (25 - 50)	59 (31 - 228)	64 (40 - 100)
Flow (L/s)	155 (51 - 344)	14 (10 - 44)	443 (110 - 700)	104 (96 - 171)

Notes:
[1] Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.

[2] Figures in brackets are annual ranges.
[3] NM indicates no measurement taken.
[4] cfu - colony forming unit.
[5] Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.
[6] Extracted from EPD River Water Quality in Hong Kong 2015

6.4.2.2 According to EPD's River Monitoring Report, Yuen Long Creek's overall compliance rate was 48% in 2015, lower than 52% in 2014. The compliance rates for upstream stations YL1 and YL2 were 62% and 60% respectively in 2015 as compared with 58% and 60% respectively in 2014. The rates for the stations in the middle of Yuen Long township (YL3 and YL4) were 33% and 35% in 2015 as compared with 57% and 33% in 2014 respectively. The river was still subject to discharges from remaining livestock farms, unsewered village houses and other specific uses (Storage and Workshop, and Open Storage) establishments.

EPD's Marine Monitoring Stations

6.4.2.3 The Pearl River carries heavy loads of suspended sediment and nutrients. The concentrations of these parameters within North Western waters are generally higher than those in the southern or eastern waters of Hong Kong in which oceanic influence is more pronounced.

6.4.2.4 The existing marine water quality can be referred to EPD's marine water quality data obtained from routine monitoring carried out at North Western WCZ (i.e. Stations NM1, NM2, NM3, NM5, NM6 and NM8) and Deep Bay WCZ (i.e. Stations DM1, DM2, DM3, DM4, DM5). The locations are shown in **Appendix 6.4**.

6.4.2.5 The latest water quality monitoring data in Year 2015 is adopted to determine the ambient water quality. Details of marine water quality monitoring data for Deep Bay WCZ and North Western WCZ are presented in **Table 6.10** and **Table 6.11** respectively. According to the Marine Water Quality in Hong Kong 2015, the Deep Bay WCZ attained an overall WQO compliance rate of 53% in 2015 as compared with 40% in 2014 while the North Western WCZ attained an overall WQO compliance rate of 67% in 2015 as compared with 61% in 2014.

Table 6.10 Summary of EPD's Routine Marine Water Quality Data for Deep Bay WCZ in Year 2015

Parameters		Deep Bay WCZ [1][2][3][4]				
		Inner Deep Bay			Outer Deep Bay	
		DM1	DM2	DM3	DM4	DM5
Temperature (°C)		25.4 (15.8 - 31.7)	25.2 (16.0 - 31.3)	25.2 (16.5 - 31.0)	25 (17.2 - 30.1)	24.7 (17.2 - 30.1)
Salinity		15.2 (4.3 - 20.9)	16.6 (4.2 - 24.8)	20.2 (8.1 - 27.9)	21.5 (9.9 - 29.7)	23.8 (7.9 - 30.8)
Dissolved Oxygen (mg/L)	Depth Average	4.5 (3.0 - 6.9)	5.1 (2.9 - 7.0)	5.5 (4.0 - 7.3)	5.8 (4.4 - 7.2)	6 (4.2 - 7.2)
	Bottom	N.M.	N.M.	N.M.	5.8 (4.2 - 7.2)	5.8 (3.8 - 7.2)
Dissolved Oxygen (% Saturation)	Depth Average	60 (39 - 83)	67 (41 - 92)	75 (56 - 103)	79 (62 - 90)	81 (60 - 91)
	Bottom	N.M.	N.M.	N.M.	78	79

Parameters	Deep Bay WCZ [1][2][3][4]				
	Inner Deep Bay			Outer Deep Bay	
	DM1	DM2	DM3	DM4	DM5
				(60 - 89)	(54 - 91)
pH	7.2 (6.8 - 7.7)	7.5 (7.3 - 7.7)	7.7 (7.4 - 7.9)	7.7 (7.6 - 7.9)	7.8 (7.6 - 8.0)
Secchi Disc Depth (m)	0.9 (0.2 - 1.5)	1 (0.7 - 2.0)	1.2 (0.7 - 2.0)	1.6 (1.0 - 2.3)	1.8 (1.3 - 2.5)
Turbidity (NTU)	27.9 (10.4 - 81.3)	20 (12.7 - 41.0)	8.9 (3.5 - 25.2)	4.9 (2.4 - 13.3)	4.2 (2.4 - 8.0)
SS (mg/L)	44.1 (13.0 - 150.0)	33.3 (14.0 - 61.0)	11.6 (4.4 - 28.0)	8 (2.7 - 15.0)	7.3 (2.7 - 18.7)
5-day Biochemical Oxygen Demand (mg/L)	3.4 (1.6 - 7.1)	2.8 (1.2 - 5.7)	1.2 (0.6 - 2.8)	0.8 (0.5 - 1.8)	0.8 (0.5 - 2.2)
Ammonia Nitrogen (mg/L)	1.34 (0.440 - 3.000)	0.913 (0.280 - 1.800)	0.256 (0.096 - 0.560)	0.202 (0.045 - 0.415)	0.158 (0.035 - 0.287)
Unionised Ammonia (mg/L)	0.016 (0.003 - 0.074)	0.019 (0.003 - 0.076)	0.006 (0.002 - 0.013)	0.005 (0.002 - 0.009)	0.005 (0.002 - 0.009)
Nitrite Nitrogen (mg/L)	0.432 (0.130 - 0.890)	0.366 (0.054 - 1.000)	0.211 (0.053 - 0.390)	0.149 (0.036 - 0.315)	0.115 (0.024 - 0.253)
Nitrate Nitrogen (mg/L)	1.069 (0.500 - 2.200)	0.94 (0.470 - 1.500)	0.943 (0.510 - 1.900)	0.903 (0.300 - 1.850)	0.795 (0.223 - 1.930)
Total Inorganic Nitrogen (mg/L)	2.84 (1.91 - 4.13)	2.22 (1.43 - 3.17)	1.41 (0.91 - 2.56)	1.25 (0.63 - 2.41)	1.07 (0.49 - 2.05)
Total Kjeldahl Nitrogen (mg/L)	2.38 (0.88 - 4.00)	1.67 (0.66 - 3.20)	0.61 (0.41 - 0.93)	0.52 (0.31 - 0.87)	0.44 (0.29 - 0.66)
Total Nitrogen (mg/L)	3.88 (2.28 - 5.13)	2.98 (1.81 - 4.67)	1.76 (1.19 - 2.94)	1.57 (1.05 - 3.01)	1.35 (0.91 - 2.43)
Orthophosphate Phosphorus (mg/L)	0.236 (0.150 - 0.310)	0.2 (0.100 - 0.350)	0.086 (0.035 - 0.130)	0.052 (0.029 - 0.074)	0.04 (0.021 - 0.067)
Total Phosphorus (mg/L)	0.37 (0.23 - 0.58)	0.3 (0.14 - 0.56)	0.16 (0.08 - 0.26)	0.11 (0.06 - 0.20)	0.09 (0.04 - 0.26)
Silica (as SiO_2) (mg/L)	8.74 (5.10 - 14.00)	7.36 (2.60 - 16.00)	4.68 (1.70 - 9.60)	4.1 (1.10 - 9.05)	3.71 (0.93 - 10.60)
Chlorophyll-a (µg/L)	15.2 (4.1 - 56.0)	12.6 (2.1 - 52.0)	4.7 (1.4 - 25.0)	2.3 (0.9 - 4.6)	2.6 (0.6 - 8.7)
***E.coli* (count/100mL)**	1800 (90 - 180000)	340 (16 - 12000)	23 (2 - 590)	27 (1 - 450)	92 (19 - 660)
Faecal Coliforms (count/100mL)	3400 (170 - 200000)	790 (62 - 20000)	54 (6 - 1000)	61 (1 - 670)	190 (50 - 1000)

Notes:
[1] Data presented are depth averaged and are the annual arithmetic mean except for *E. Coli* (geometric mean)
[2] Data in brackets indicate ranges
[3] Extracted from EPD Marine Water Quality in Hong Kong 2015
[4] N.M. – not measured.

Table 6.11 Summary of EPD's Routine Marine Water Quality Data for North Western WCZ in Year 2015

Parameters		North Western WCZ [1][2][3]					
		NM1	NM2	NM3	NM5	NM6	NM8
Temperature (°C)		23.7 (16.9 - 27.4)	24 (16.9 - 27.8)	24.2 (17.0 - 27.7)	24.3 (16.9 - 28.0)	24.4 (16.6 - 28.7)	24.2 (16.6 - 28.6)
Salinity		29.9 (25.2 - 31.7)	28.1 (18.9 - 31.3)	27.9 (21.0 - 31.3)	26.3 (21.2 - 31.0)	25.5 (14.3 - 31.4)	27.8 (14.0 - 32.8)
Dissolved Oxygen (mg/L)	Depth Average	5.4 (3.7 - 7.2)	5.7 (4.3 - 7.4)	5.6 (4.3 - 7.3)	5.7 (4.1 - 7.5)	6.1 (4.9 - 7.9)	6.2 (4.8 - 8.2)
	Bottom	5.2 (2.5 - 7.3)	5.4 (3.2 - 7.3)	5.3 (3.0 - 7.4)	5.4 (2.9 - 7.5)	5.9 (3.9 - 8.0)	6 (3.3 - 8.3)
Dissolved Oxygen (% Saturation)	Depth Average	75 (53 - 90)	79 (62 - 92)	78 (62 - 91)	78 (57 - 93)	84 (70 - 98)	87 (70 - 102)
	Bottom	72 (36 - 92)	75 (46 - 91)	74 (44 - 91)	74 (42 - 93)	81 (56 - 99)	83 (48 - 103)
pH		7.8 (7.7 - 8.0)	7.8 (7.6 - 8.0)	7.8 (7.6 - 8.0)	7.8 (7.6 - 8.0)	7.8 (7.6 - 8.0)	7.9 (7.7 - 8.1)
Secchi Disc Depth (m)		2.5 (1.6 - 3.5)	2.4 (1.3 - 3.8)	2.2 (1.5 - 3.0)	2.2 (1.2 - 3.0)	2.1 (1.0 - 3.0)	1.9 (1.1 - 2.5)
Turbidity (NTU)		6 (3.3 - 10.5)	5.5 (2.9 - 11.4)	7.5 (3.2 - 13.9)	6.7 (3.7 - 9.3)	6.8 (4.0 - 9.9)	8.5 (5.6 - 13.8)
SS (mg/L)		8 (1.3 - 22.0)	6.6 (2.7 - 20.3)	10.8 (4.2 - 30.3)	8.6 (4.5 - 20.4)	9.2 (2.4 - 18.3)	12.8 (3.5 - 22.3)
5-day Biochemical Oxygen Demand (mg/L)		0.6 (<0.1 - 2.6)	0.8 (<0.1 - 2.8)	0.8 (<0.1 - 2.8)	0.6 (<0.1 - 2.2)	0.8 (<0.1 - 3.0)	0.7 (<0.1 - 1.8)
Ammonia Nitrogen (mg/L)		0.107 (0.016 - 0.177)	0.106 (0.014 - 0.180)	0.11 (0.013 - 0.203)	0.115 (0.013 - 0.207)	0.087 (0.011 - 0.177)	0.047 (0.020 - 0.140)
Unionised Ammonia (mg/L)		0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.009)	0.003 (<0.001 - 0.008)	0.003 (<0.001 - 0.009)	0.002 (<0.001 - 0.009)	0.002 (<0.001 - 0.008)
Nitrite Nitrogen (mg/L)		0.052 (0.016 - 0.130)	0.066 (0.017 - 0.131)	0.07 (0.018 - 0.153)	0.091 (0.020 - 0.270)	0.082 (0.014 - 0.203)	0.056 (0.007 - 0.140)
Nitrate Nitrogen (mg/L)		0.286 (0.127 - 0.723)	0.421 (0.133 - 1.230)	0.429 (0.137 - 0.963)	0.557 (0.153 - 1.090)	0.597 (0.130 - 1.430)	0.406 (0.026 - 1.600)
Total Inorganic Nitrogen (mg/L)		0.44 (0.27 - 0.84)	0.59 (0.29 - 1.36)	0.61 (0.31 - 1.09)	0.76 (0.33 - 1.23)	0.77 (0.27 - 1.55)	0.51 (0.09 - 1.69)
Total Kjeldahl Nitrogen (mg/L)		0.37 (0.18 - 0.59)	0.37 (0.15 - 0.69)	0.38 (0.19 - 0.80)	0.38 (0.19 - 0.60)	0.33 (0.18 - 0.51)	0.29 (0.12 - 0.52)
Total Nitrogen (mg/L)		0.71 (0.56 - 1.06)	0.86 (0.52 - 1.61)	0.88 (0.55 - 1.28)	1.03 (0.64 - 1.46)	1.01 (0.50 - 1.86)	0.75 (0.30 - 2.01)
		0.022	0.024	0.025	0.028	0.025	0.017

Parameters	North Western WCZ [1][2][3]					
	NM1	NM2	NM3	NM5	NM6	NM8
Orthophosphate Phosphorus (mg/L)	(0.008 - 0.031)	(0.013 - 0.039)	(0.011 - 0.040)	(0.011 - 0.047)	(0.007 - 0.050)	(0.006 - 0.038)
Total Phosphorus (mg/L)	0.05 (0.03 - 0.07)	0.05 (0.03 - 0.08)	0.06 (0.04 - 0.09)	0.06 (0.03 - 0.10)	0.06 (0.03 - 0.11)	0.05 (<0.02 - 0.10)
Silica (as SiO_2) (mg/L)	1.58 (0.24 - 3.83)	2.18 (0.27 - 6.87)	2.17 (0.29 - 5.30)	2.7 (0.39 - 6.03)	2.86 (0.57 - 7.77)	2.24 (0.25 - 8.80)
Chlorophyll-a (µg/L)	1.6 (0.4 - 4.6)	1.8 (0.4 - 5.8)	1.7 (0.7 - 4.1)	1.5 (0.5 - 2.9)	1.9 (0.9 - 4.4)	2.4 (0.7 - 6.2)
***E.coli* (count/100mL)**	97 (7 - 510)	31 (2 - 220)	59 (10 - 220)	130 (42 - 1000)	19 (7 - 140)	3 (<1 - 27)
Faecal Coliforms (count/100mL)	200 (48 - 650)	70 (4 - 340)	130 (23 - 580)	260 (89 - 2900)	43 (17 - 210)	6 (1 - 53)

Notes:
[1] Data presented are depth averaged and are the annual arithmetic mean except for *E. Coli* (geometric mean)
[2] Data in brackets indicate ranges
[3] Extracted from EPD Marine Water Quality in Hong Kong 2015

6.4.2.6 The Deep Bay is affected by the discharge from the Pearl River, which contains wastewater with high level of nutrients. Since the North Western WCZ is adjacent to the Deep Bay WCZ, the North Western WCZ also has relatively high total inorganic nitrogen due to the influence of the Pearl River and the Deep Bay.

Baseline Water Quality Monitoring at Existing Watercourses

6.4.2.7 Other than the desktop information as summarised above, in-situ water quality monitoring has been conducted at selected watercourses within the PDA. Monitoring locations are selected in consideration of accessibility. Access to channelised watercourses are generally restricted. There are a total of 14 watercourses within the PDA. Out of these watercourse, 5 watercourses (B1-B5) are accessible. These accessible watercourses are distributed throughout the PDA relatively evenly. It is thus anticipated that the water quality in these watercourses can in general reflect the water quality of the other watercourses within the PDA. In addition, 2 channelised watercourses downstream of the PDA (B6-B7) which currently do not have baseline monitoring data are also selected for monitoring. Locations of the monitoring points are shown in **Figure 6.1**. The following table summarises the monitoring results.

Table 6.12 Baseline Water Quality Monitoring at Existing Watercourses under this Study

Monitoring Locations	Watercourse	Temperature (°C)	pH	Dissolved oxygen DO (mg/L)
B1	S20	25.0	6.2	3.30
B2	S02	24.3	6.5.	1.45
B3	S07	24.3	6.9	2.50
B4	S26	25.6	7.0	5.32
B5	S04	23.9	5.9	1.27
B6	S17	30.0	7.3	4.37

Monitoring Locations	Watercourse	Temperature (°C)	pH	Dissolved oxygen DO (mg/L)
B7	S18	27.6	7.2	6.82

6.4.2.8 Baseline water quality monitoring at several existing watercourses within and outside the PDA were conducted in May 2017 and July 2017 respectively to supplement the water quality data. The measured water temperature range from 23.9 – 30.0 and the pH range from 5.9 – 7.3. The DO levels range from 1.45 – 6.82 mg/L, with data measured in B1 – B3 and B5 below 4 mg/L. The low DO levels may be due to discharge from unsewered population nearby. Based on the sampling results of these rivers, it can be shown that the watercourses within the PDA are generally subject to disturbance by human activities.

6.5 Assessment Methodology

6.5.1.1 In accordance with the EIA Study Brief, the assessment area for water quality impact assessment includes the area within 500m from the Project Site boundary. The assessment area would be extended to include other areas such as watercourses, fish ponds in the vicinity that may be impacted.

6.5.1.2 The major area of concern during construction and operation of the Project are the sewage from workforce and non-point sources discharge such as additional surface runoff due to change of hydrology regime. There will be neither dredging, nor reclamation works and all the works will be land-based. Hence, water quality modelling is not proposed.

6.5.1.3 The assessment approach is referred to Annex 6 – Criteria for Evaluating Water Pollution and Annex 14 – Guidelines for Assessment of Water Pollution under the EIAO-TM.

6.6 Identification of Pollution Sources

6.6.1.1 The proposed development comprises the following works which may have impact on water quality. Detailed description of the works are presented in **Chapter 3**. Some of the development would also fall under Schedule 2 Designated Projects (DPs). These DPs are described in details in **Section 1.5**. The potential water quality impact during construction and operational phases are summarised in **Table 6.13** and **Table 6.14**.

- Site formation, including removal/ diversion of watercourses and removal of ponds;
- Road works and utilities;
- Bridge structures, underpass at interchanges and bridge crossing over channelised watercourse;
- Construction of superstructures for residential, commercial uses, other associated facilities and storage and workshops;
- STW/ Sewage Pumping Stations(SPSs)/ Reedbed/ Reclaimed Water Service Reservoir;

- Provision of a new sewer from YLS STW to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall;
- Revitalisation of Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah; and
- Construction of box culverts.

6.6.1.2 The pollution sources/impacts for the proposed developments, including both DP and non-DP elements during the construction phase are summarized as follows.

- Site Run-off from general site operation;
- Accidental spillage of chemicals;
- Sewage from workforce;
- Contaminated groundwater and wastewater;
- Construction run-off from works near watercourses;
- Run-off during removal/ diversion of watercourses;
- Run-off during removal / filling of ponds;
- Run-off during decking over sections of Yuen Long Nullah along Kung Um Road and Kiu Hing Road;
- Run-off during revitalisation of Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah;
- Potential pollution due to construction of box culvert.

Table 6.13 Potential water quality impact caused by construction of DPs

Potential impacts	Designated Projects (see Figure 1.2 for locations)									
	DP1	DP2	DP3	DP4[2]	DP5	DP6	DP7[2]	DP8	DP9[2]	DP10
Site Run-off from general site operation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Accidental spillage of chemicals	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Sewage from workforce	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Contaminated groundwater and wastewater [3]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Construction Works near Watercourses	✓	✓	✓	✓	✓		✓	✓	✓	✓
Decking over sections of Yuen Long Nullah along Kung Um Road and Kiu Hing Road										✓
Revitalisation of Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah										✓

Potential impacts	Designated Projects (see Figure 1.2 for locations)									
	DP1	DP2	DP3	DP4[2]	DP5	DP6	DP7[2]	DP8	DP9[2]	DP10
Potential pollution due to construction of box culvert										✓

Note:
[1] DP1: Trunk Roads (TSW West Interchange)
DP2: New Primary Distributor Roads (TYST Interchange)
DP3: Two new Distributor Roads (Road D1 to D2)
DP4: Environmentally Friendly Transport Services (EFTS)
DP5: Slip roads at the TYST Interchange
DP6: Partly depressed road/ underpass located at TSWW Interchange and full enclosures at TYST Interchange
DP7: New YLS STW
DP8: Two new SPSs
DP9: Reclaimed Water Service Reservoir for Reuse of Reclaimed Water
DP10: Yuen Long Nullah revitalisation/decking along Kung Um Road and Kiu Hing Road
[2] For DP4, DP7 and DP9, the specific construction works will not be conducted under this Project, hence only the impacts due to site formation are relevant.
[3] The occurrence of contaminated groundwater is to be determined upon land contamination assessment results after land resumption. The general impact and mitigation measures associated with contaminated groundwater is included in **Sections 6.7.1.7** and **6.8.1.9**.

6.6.1.3 The pollution sources/impacts for the proposed developments, including both DP and non-DP elements during the operational phase are summarized as follows.

- Sewage disposal;
- Discharge to the Deep Bay WCZ;
- Discharge to the North Western WCZ;
- Surface run-off from the PDA;
- Emergency discharge from STW;
- Emergency discharge from SPSs;
- Maintenance flushing for Reclaimed Water Service Reservoir for reuse of reclaimed water; and
- Wastewater from industrial and commercial activities.

Table 6.14 Potential water quality impact caused by DPs and other development works during operational phase

Potential impacts	Designated Projects (see Figure 1.2 for locations)									
	DP1	DP2	DP3	DP4[2]	DP5	DP6	DP7[2]	DP8	DP9[2]	DP10
Sewage disposal							✓	✓	✓	
Discharge to the Deep Bay WCZ							✓			
Discharge to the North Western WCZ							✓			
Surface run-off from the PDA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Potential impacts	Designated Projects (see Figure 1.2 for locations)									
	DP1	DP2	DP3	DP4[2]	DP5	DP6	DP7[2]	DP8	DP9[2]	DP10
Emergency discharge from Sewage Treatment Works							✓			
Emergency discharge from Sewage Pumping Stations								✓		
Maintenance flushing for Reclaimed Water Service Reservoir for reuse of reclaimed water									✓	

Note:
[1] DP1: Trunk Roads (TSW West Interchange)
DP2: New Primary Distributor Roads (TYST Interchange)
DP3: Two new Distributor Roads (Road D1 to D2)
DP4: Environmentally Friendly Transport Services (EFTS)
DP5: Slip roads at the TYST Interchange
DP6: Partly depressed road/ underpass located at TSWW Interchange and full enclosures at TYST Interchange
DP7: New YLS STW
DP8: Two new SPSs
DP9: Reclaimed Water Service Reservoir for Reuse of Reclaimed Water
DP10: Yuen Long Nullah revitalisation/decking along Kung Um Road and Kiu Hing Road
[2] DP4, DP7 and DP9 will be studied and implemented under separated EIA(s). The cumulative impacts due to these DPs are assessed in this EIA.

6.7 Prediction and Evaluation of Impacts

6.7.1 Construction Phase

6.7.1.1 During the construction phase, general construction works including site formation, construction of roads and utilities, infrastructures, superstructures for residential and commercial development, sewage treatment facilities (i.e. SPSs and reedbed), the new sewer from YLS STW to existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall, and works within / near to watercourses may lead to various potential water quality impacts. These impacts include site run-off, accidental chemical spillage, pollution from workforce sewage, contamination groundwater etc. Construction of infrastructures associated with existing watercourse, such as reedbed/retention pond, hillside river corridor/retention lake, which may lead to pollution if the works are not properly contained and isolated from the existing watercourse. These potential impacts are assessed in the following sub-sections. The construction of EFTS (DP4), STW (DP7) and Reclaimed Water Service Reservoir (DP9) will be implemented by other studies. It is anticipated that general construction activities such as site formation and superstructure works would be involved, and potential impacts include site run-off, accidental chemical spillage, pollution from workforce sewage etc. will be considered as cumulative impacts (refer to **Section 6.7.3**).

Site Run-off from General Site Operation

6.7.1.2 During rainstorm events, construction site run-off would come from all over the works site. The surface runoff might be polluted by:

- Runoff and erosion from site surfaces, earth working areas, etc.;
- Wash water from dust suppression sprays and wheel washing facilities; and
- Bentonite slurries for foundation works.

6.7.1.3 Construction runoff may cause physical, biological and chemical effects. The physical effects include potential blockage of drainage channels and increase of suspended solid (SS) levels in the Deep Bay WCZ. Runoff containing significant amounts of concrete and cement-derived material may cause primary chemical effects such as increasing turbidity and discoloration, elevation in pH, and accretion of solids. A number of secondary effects may also result in toxic effects to water biota due to elevated pH values, and reduced decay rates of faecal micro-organisms and photosynthetic rate due to the decreased light penetration. Appropriate precautionary measures shall be adopted to prevent site run-off to the watercourses retained within and in the vicinity of the Project Boundary.

6.7.1.4 An Ecologically Important Stream (EIS) in Yeung Ka Tsuen is identified at around 100m to the southeast of the Area 3 PDA. As discussed in **Section 1.7**, during the earlier planning stage, the boundaries of the PDA are designed to avoid encroachment onto the EIS near Yeung Ka Tsuen, which is now located outside the PDA boundary. In addition, the Study has proposed rezoning of these areas, so that the ecologically important resources within these "Undetermined" land uses can be conserved. Besides, the water quality in this EIS is good and hence any polluted site run-off shall be avoided from flowing into the EIS. Given a 100m buffer distance of the EIS has been allowed in the RODP, potential water quality impact due to site run-off would be low. Appropriate precautionary measures should still be implemented to control site run-off. With the implementation of measures stated in **Section 6.8.1**, adverse water quality impact to the EIS is not anticipated.

Accidental Spillage of Chemicals

6.7.1.5 The chemicals used during construction, such as fuel, oil, solvents and lubricants from maintenance of construction machinery and equipment, may cause pollution and trigger physicochemical effects in the nearby water bodies if accidental spillage occurs. To avoid adverse impacts of chemical spillage, best practices of chemical storage practices such as storage under covered area, provision of secondary containment and material safety data sheets are advised. Spill kits are also advised to handle spillage and the staff should be trained for handling spillage. With the implementation of mitigation measures stated in **Section 6.8.1**, adverse water quality impact to the watercourses and EIS is not anticipated.

Sewage from Workforce

6.7.1.6 Sewage effluents will arise from the sanitary facilities provided for the on-site construction workforce. According to Table T-2 of Guidelines for Estimating Sewage Flows for Sewage Infrastructure Planning, the unit flow is 0.23 m^3/day/employed population. The characteristics of sewage would include high levels of BOD_5, Ammonia and *E. coli* counts. Since portable chemical toilets and sewage holding tank will be provided, no adverse water quality impact is anticipated.

Contaminated Groundwater and Wastewater

6.7.1.7 If excavations for the Project take place below the groundwater table, there may be a need to dewater the pits for safety and construction purposes. Where dewatering takes place through layers of contaminated material or where any contaminated soil is being excavated, the groundwater may then become contaminated, thereby

requiring appropriate handling and disposal. Besides, contaminated wastewater and runoff will also be generated from the excavated contaminated soil during handling works and precipitation. All the contaminated water should be collected, treated and disposed in a manner as listed in **Section 6.8.1.9**. Hence, with such mitigation measures in place, no significant water quality impact due to the contaminated water is expected.

Construction Run-off from Works near Watercourses

6.7.1.8 Some of the watercourses are located within the Project area as identified in **Section 6.3**. Construction works, including those for the DPs, near these watercourses may pollute the storm water or inland waters due to potential release of construction wastes. Construction wastewater are generally characterised by high concentration of SS and elevated pH. Adoption of good housekeeping and mitigation measures would reduce the generation of construction wastes and potential water pollution. The implementation of measures to control run-off and drainage water will be important for the construction works adjacent to the inland water in order to prevent run-off and drainage water with high levels of SS from entering the water environment. With the implementation of adequate construction site drainage and BMPs as described in **Section 6.8.1** and provision of mitigation measures as specified in ETWB TC (Works) No. 5/2005 "Protection of natural streams/rivers from adverse impacts arising from construction works", it is anticipated that water quality impacts would be minimised.

Water pollution during Removal/ Diversion of Watercourses

6.7.1.9 The major channelised watercourse including Yuen Long West Nullah, Tin Tsuen Channel and Yuen Long Nullah will be retained and revitalised. As indicated in **Table 6.8** and **Figure 6.1**, watercourses S02, S03 and S04 would be preserved. The lower section of S02 will be diverted to S03 and eventually flow into the Yuen Long (West) Nullah. Some watercourses within the PDA will be permanently lost (refer to **Table 6.8)**. Also, the construction of infrastructures associated with the existing watercourses, such as reedbed/retention pond, hillside river corridor/retention lake will alter the flow of the existing watercourses.

6.7.1.10 The diversion of watercourses would involve excavation and construction works. If not properly controlled, the excavated materials, waste water, chemicals or other construction materials may enter the watercourses and give rise to water quality impact at the downstream area. To prevent adverse water quality impact, the stream diversion works should schedule in dry season as far as practicable when the flow is low. Sequencing of works should be duly planned to minimise water quality impacts. A temporary drainage or permanent drainage at most downstream should first be established to intercept and divert the upstream flow. Precautionary measures in Appendix D of ETWB TC No.5/2005, such as temporary isolation to natural rivers using sandbags and silt curtains, shall be applied to prevent pollutants running into the neighbouring watercourse. Dewatering of the construction works area shall be conducted prior to the construction works. Silt removal facilities should be adopted to treat the wastewater from dewatering operation prior to discharge. After completion of the construction works, the works area shall be cleaned up before receiving any water flow or connecting to any existing watercourses. Detailed mitigation measures are given in **Section 6.8.1**.

6.7.1.11 For the watercourses within the PDA that will be filled, the same approach in controlling potential water pollution, including diversion of waterflow, isolation of workfronts, dewatering of works area and proper clean up upon completion of work

shall be followed. Small amount of land-based sediment may be involved when removing the bedding sediment of the streams before filling. The management of land-based sediment will be discussed in **Section 11.5.1**.

Run-off during Removal/ Filling of Ponds

6.7.1.12 As indicated in **Table 6.8**, there are 9 ponds within the PDA. These ponds identified within the PDA are for non-aquaculture use, and will be filled for the proposed development. These ponds are fragmented and are not hydrologically linked to other watercourses. In review of the inactive and fragmented status, potential impacts to the affected ponds would be insignificant.

6.7.1.13 These ponds will be dewatered and the sediment will be removed from the ponds. Proper management of the drained water and sediment shall be implemented, to prevent release to existing watercourse. The management of land-based sediment due to the filling of ponds will be discussed in **Section 11.5.1**.

Run-off during Decking over sections of Yuen Long Nullah along Kung Um Road and Kiu Hing Road

6.7.1.14 Pollutants such as dust and construction materials from the works over Yuen Long Nullah may fall into the nullah and increase the SS level in the water. Mitigation measures as specified in ETWB TC (Works) No. 5/2005 "Protection of natural streams/rivers from adverse impacts arising from construction works" shall be implemented. Also, toe boards or bunds should be provided along the works platform over the nullah to prevent the construction materials or debris from falling into the nullah. Any stockpiling of dusty materials shall be placed away from the nullah. Detailed mitigation measures are given in **Section 6.8.1**. With the mitigation measures implemented, adverse water quality impact would be minimal.

Run-off during Revitalisation of Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah

6.7.1.15 Revitalisation of the watercourses would involve minor excavation and laying of river bank, and may have works area within existing watercourses. Run-off and release of pollutants into the channels will increase the SS level in the water. Also, temporary drainage diversion would be required during the construction works. The works within the nullah shall be isolated from the flow of the nullah by installation of cofferdams and temporary work platform.

6.7.1.16 The Contractor should provide the detailed design of the cofferdams in accordance with the DSD Technical Circular No. 1/2017 "Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater Drainage System" and DSD Practice Note No. 1/2004 "Design Rainfall Depth for Temporary Works within the Dry Season" for DSD approval in order to finalize the options of these temporary structure before the commencement of works, to avoid adverse impact to the drainage characteristics of the nullah. Detailed mitigation measures are given in **Section 6.8**. With the mitigation measures implemented, adverse water quality impact would be minimal.

Potential Pollution due to Construction of Box Culvert

6.7.1.17 Construction of box culvert will involve excavation, which may lead to temporary increase of SS level and turbidity. Also, construction works may cause run-off and release of pollutants into the channels. Flow diversion should be conducted before the construction of box culvert. The works within the nullah shall be isolated from the flow of the nullah by installation of cofferdams and temporary work platform.

Moreover, a detail design of the cofferdams should be prepared by the Contractor as mentioned in **Section 6.7.1.16**.

6.7.2 Operational Phase

Sewage Disposal

6.7.2.1 During the operational phase, sewage discharge will be the major water pollution source. These would include domestic sewage from residential and commercial land uses and wastewater from other specific uses such as Storage and Workshop, and Open Storage.

6.7.2.2 To address the sewage generated from the development, a comprehensive sewage treatment and disposal strategy is formulated and detailed in **Section 7.6.3**. In summary, a new STW will be proposed and designed to achieve a total treatment capacity of 24,000m^3/day in ADWF and up to tertiary level treatment standard to enable for reuse of treated sewage effluent (TSE) as reclaimed water while the rest of sewage is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to the Urmston Road submarine outfall. For the reclaimed water, it will be reused for non-potable uses such as toilet flushing in the PDA and adjacent developments; and only a small amount of TSE will be discharged to the reedbed adjoining to the proposed YLS STW for further polishing before entering Yuen Long Nullah.

6.7.2.3 The project proponent would further study the feasibility of reusing the reclaimed water from the Reclaimed Water Service Reservoir for toilet flushing in Tin Shui Wai Area.

Discharge to the Deep Bay WCZ

Improvement of Pollution Loading from Livestock Waste

6.7.2.4 As mentioned in **Section 1.7**, there are currently 3 pig farms and 3 chicken farms in the PDA whose effluent would be discharged into nearby watercourses. Release of livestock waste into nearby watercourse would increase the nutrient level in the water environment, deteriorating the water quality of these watercourses. However, as 3 pig farms and 2 chicken farms will be decommissioned, the pollution loading to these watercourse and the Deep Bay WCZ would be significantly reduced after the proposed development. This would be recognized as the pollution credit gained from this Project. The existing and likely future abated pollution loadings are shown in **Table 6.15.** Details of the calculation are shown in **Appendix 6.1**.

Table 6.15 Improvement of Pollution Loading from Livestock Farm

Parameters	Unit	Existing Farms & Likely Future Farms (Without Project)	Likely Future Farms (With Project)	Improvement (Without vs. With Project)
Flow	m3/d	117.03	4.55	96%
BOD_5	kg/d	5.85	0.23	96%
SS	kg/d	5.85	0.23	96%
NH_3-N	kg/d	109.50	10.50	90%
OrgN	kg/d	102.40	10.50	90%
TIN [1]	kg/d	109.50	10.50	90%
TN [2]	kg/d	211.90	21.00	90%
TP	kg/d	95.30	10.50	89%

Parameters	Unit	Existing Farms & Likely Future Farms (Without Project)	Likely Future Farms (With Project)	Improvement (Without vs. With Project)
E. coli	count/d	4.32E+15	2.12E+14	95%

Notes:
[1] TIN is assumed to be equal to NH_3-N only as a conservative approach which will underestimate the improvement.
[2] TN is assumed to be equal to NH_3-N plus OrgN only as a conservative approach which will underestimate the improvement.

Improvement of Pollution Loading from Unsewered Population

6.7.2.5 As discussed in **Section 1.7**, there is no sewerage within the PDA, and the watercourses are subject to discharges from the unsewered population. The estimated existing unsewered population within the scattered villages and brownfield sites within the PDA approximately 280 (about 100 households), and the estimated employment is about 4,500 within the existing brownfield sites within the PDA. With the proposed sewage treatment plant implemented under the Project, all the areas within the PDA boundary will be provided with sewers that convey the sewage to the new sewage treatment plant. Hence, all of the unsewered population within the PDA will be sewered and the pollution loading from the unsewered population will be abated. The credit of pollution loading gained from providing sewerage to the existing population is shown in **Table 6.16**. Detailed calculation is shown in **Appendix 6.2**.

Table 6.16 Improvement of Pollution Loading of Unsewered Population

Parameters	Unit	Existing Case	Likely Future Case (Without project)	Likely Future Case (With project)	Improvement (Without vs. With Project)
Flow	m3/d	1555	1868	0	100%
BOD_5	kg/d	222	266	0	100%
SS	kg/d	83	100	0	100%
NH_3-N	kg/d	69	83	0	100%
OrgN	kg/d	[1]	[1]	[1]	-
TN	kg/d	69	83	0	100%
TP	kg/d	7.5	9.0	0	100%
E. coli	count/d	1.70E+11	2.04E+11	0	100%

Notes:
[1] According to EIA-190/2010 for "Liantang / Heung Yuen Wai Boundary Control Point and Associated Works", most of the organic nitrogen would be converted to NH3-N by microorganism in septic tank. As such, organic nitrogen is assumed to be negligible and total nitrgen is assumed to be equivalent to NH3-N.

Improvement of Pollution Loading from Surface Run-off

6.7.2.6 As discussed in **Section 6.7.2.11** to **Section 6.7.2.13**, the pollution loading from surface run-off would drop after the development as a result of the incorporation of silt trap. **Table 6.19** summarizes the pollution loading before and after the development.

Sewage Pollution Loading from the PDA

6.7.2.7 As discussed in **Section 7.6.3**, a small amount of TSE of about 570 $m^{3/}$day which is treated by tertiary treatment would be discharged to a reedbed before entering Yuen Long Nullah. The pollution loading is shown in **Table 6.17**. It should be noted that

the extra polishing effect of the reedbed is not taken into account as a conservative approach.

Table 6.17 Pollution Loading of TSE of about 570 $m^{3/}$day

Parameters [1]	Unit	TSE discharge to Yuen Long Nullah
Flow	m3/d	570
BOD_5	kg/d	5.62
SS	kg/d	2.78
NH_3-N	kg/d	0.57
OrgN	kg/d	1.14
TIN	kg/d	4.56
TN	kg/d	5.70
TP	kg/d	1.29
E. coli	count/d	Not Detectable

Notes:
[1] The standard of the TSE (i.e. tertiary treatment) is referenced to **Section 7**. The other parameters OrgN, TIN, TN and TP which are not regulated in the aforementioned standard are assumed to be 2mg/L, 8mg/L, 10mg/L and 2.26mg/L respectively following the standard of secondary plus treatment from HSK EIA as a conservative approach.

Overall Improvement

6.7.2.8 As shown in **Sections 6.7.2.4** to **Section 6.7.2.7**, most of the pollution loading would either be completely removed (e.g. livestock farms and unsewered population) or would decline as a result of the implementation of mitigation measures (e.g. surface run-off). The only introduction of pollution loading would be the TSE of about 570 $m^{3/}$day, treated by tertiary treatment. As a conservative approach, the pollution loading of the TSE is fully compensated by only taking into account the pollution credits gained from the abatement of livestock farm, as shown in **Table 6.18**. Therefore, it is anticipated that there would be no net increase in the pollution loading to the Deep Bay WCZ, complying with the requirement of "No Net Increase in Pollution Loads Requirement in Deep Bay".

Table 6.18 Overall Improvement of Pollution Loading

Parameters	Unit	TSE discharge to Yuen Long Nullah	Decrease in Pollution Loading from Livestock	Overall Improvement [1]
Flow	m3/d	570	112.48	-
BOD_5	kg/d	5.62	5.62	0.00
SS	kg/d	2.78	5.62	2.84
NH_3-N	kg/d	0.57	99.00	98.43
OrgN	kg/d	1.14	91.90	90.76
TIN	kg/d	4.56	99.00	94.44
TN	kg/d	5.70	190.90	185.20
TP	kg/d	1.29	84.80	83.51
E. coli	count/d	Not Detectable	4.11E+15	4.11E+15

Notes:
[1] The improvement only take into account the abatement of livestock farm as a conservative approach.

Discharge to the North Western WCZ

6.7.2.9 As discussed in **Section 6.7.2.2**, part of the sewage will be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to the

Urmston Road submarine outfall which is located in the North Western WCZ. **Table 7.9** summarizes the pollution loading of the total combined effluent at the Urmston Road submarine outfall from SW STW, HSK STW and YLS STW against the pollution loading approved in the *Upgrading and expansion of San Wai Sewage Treatment Works and Expansion of Ha Tsuen Pumping Station EIA Report* (EIA-086/2002) ("SWSTW EIA"). The loading from the total combined effluent is smaller than that was approved in SW STW EIA for all parameters including *E. coli*, TSS, BOD5, ammonia nitrogen and total inorganic nitrogen. So it is anticipated that the potential water quality impact of the sewage effluent would not be worse than that identified in SWSTW EIA.

6.7.2.10 The proposed YLS STW is a DP under Schedule 2 DPs requiring EPs. Hence, a separate EIA Study for proposed STW will be conducted by the future project proponent.

Surface run-off from the PDA

6.7.2.11 The existing area in the vicinity is partially rural area. With the development of the Project, there would be an increase in the total paved area. Such change of pavement will reduce the infiltration rate in the catchment, consequently resulting in a higher flood risk as extra surface run-off may be generated during rain events. The flood risks would be assessed and addressed separately in the Drainage Impact Assessment Report under this study. Subject to the requirement of relevant government departments, it is recommended that the capacities of road drainage system shall cater for the runoff from 50 year-return-period rainstorm.

6.7.2.12 In terms of water quality impact, there would be pollution loading in association with the surface runoff, which is known as non-point source pollutions during operational phase. Substances such as vehicle dust, tyre scraps and oils deposited and accumulated on the road surfaces will be washed into nearby drainage system or watercourses during rainfall events. Under normal condition, runoff will not be generated in low rainfall intensity. However, the worst scenario to water quality will take place during the first flush under heavy rainstorm events. Nevertheless, proper drainage systems with silt traps should be installed and the design of road gullies with silt traps should be incorporated in later detailed design.

6.7.2.13 The total loading of non-point source pollution due to the development is compared with that of existing condition in **Appendix 6.3**. **Table 6.19** summarises the pollution loading from the existing and likely future surface run-off. The pollution loading will decline because of the incorporation of silt trap in the proposed development.

Table 6.19 Pollution Loading of Surface Run-off

Parameters [1]	Unit	Existing Case and Likely Future Case (Without Project)	Likely Future Case (With Project)	Improvement (Without vs. With Project)
BOD_5	kg/d	156.5	128.8	18%
SS	kg/d	301.1	247.8	18%
NH_3-N	kg/d	1.4	1.1	18%
OrgN	kg/d	8.4	6.9	18%
TIN	kg/d	4.2	3.4	18%
TN	kg/d	12.5	10.3	18%
TP	kg/d	0.3	0.2	18%

Notes:

[1] OrgN is equal to TKN minus NH3N, TIN is equal to NH3N plus TON. TN is equal to TKN plus TON.

Emergency discharge from Sewage Treatment Works

6.7.2.14 To prevent the occurrence of emergency discharge, contingency measures such as twin rising mains, standby pump and treatment facilities and dual electricity supply or backup power supply facilities would be provided to SPSs and STW as recommended in **Section 7.6.5**. Details of the mitigation measures are discussed in **Section 6.8.2.1**. With the above design provision as contingency measures, the risk of failure of YLS STW is considered to be negligible. In case of an extremely unlikely situation that an emergency discharge has occurred, the discharge from the proposed YLS STW will be discharged to the nearby proposed reedbed (as shown in **Figure 7.1**) which can further polish the effluent. The effluent will then be delivered to Deep Bay. Details of the emergency discharge routes under different scenario are listed in **Section 7.6.5**.

6.7.2.15 The proposed YLS STW is a DP under Schedule 2 DPs requiring EPs. Hence, a separate EIA Study for proposed STW will be conducted by the future project proponent. The exact location of the emergency discharge will also be subject to detailed design and the separate EIA Study will ensure the discharge locations would not cause significant water quality impact to the receiving water bodies.

Emergency discharge from Sewage Pumping Stations

6.7.2.16 There will be two proposed SPSs within the PDA – one (SPS1) is near the east of TYST Interchange in Area 1 and the other (SPS2) is near the west of the northern end of Kung Um Road in Area 2. The design capacities of these SPSs are shown in **Table 6.20**. To prevent the occurrence of emergency discharge, contingency measures such as twin rising mains, standby pump and treatment facilities and dual electricity supply or backup power supply facilities would be provided to SPSs and STW as recommended in **Section 7.6.5**. Details of the mitigation measures are discussed in **Section 6.8.2.1**. With the above design provision as contingency measures, the risk of failure of SPS is considered to be negligible. In case of an extremely unlikely situation that an emergency discharge has occurred, the discharge from the proposed SPS1 and SPS2 will be delivered to the nullahs near Shan Ha Road and Kiu Hing Road respectively which would finally discharge to the Deep Bay. These two nullahs are part of the channelised Yuen Long Nullah with limited ecological value. Given that the emergency discharge is very rare and these two nullahs are of limited ecological value, the water quality impact is anticipated to be short-term and insignificant.

Table 6.20 Design Capacity of SPS

SPS	Design Capacity (m3/day)
SPS1	4,700
SPS2	23,280

6.7.2.17 For the surplus TSE export disposal strategy as discussed in **Section 7.6.3**, the new sewer from YLS STW to existing inlet chamber of the NWNT effluent tunnel for discharging to the Urmston Road submarine outfall would be a gravity sewer and hence does not require any SPSs. So the emergency discharge of the SPS as a result of this export scheme is not anticipated.

Maintenance Flushing for Reclaimed Water Service Reservoir for Reuse of Reclaimed Water

6.7.2.18 The proposed Reclaimed Water Service Reservoir for reuse of TSE as reclaimed water would be located at the southwest hillside of PDA Area 3. It will be covered and this will prevent overflow during rainstorm events. Under normal operation, there is no discharge from the service reservoir to the nearby water body. As part of the necessary maintenance tasks, the reservoir would need to be flushed for cleaning but the frequency is not high. Before the flushing, the reclaimed water inside the reservoir would be discharged to the foul sewers and the sludge remained at the bottom will be collected and disposed by a licensed waste collector. During the flushing, the cleansing water would be collected by a licensed waste collector via the washout pipes at a controlled manner to avoid the possible impact to the nearby streams with ecological importance. Similar procedures are applicable to the proposed combined flushing water service reservoir and fresh water service reservoir of HSK. Given no flushing water to be discharged into the catchment, it is considered that the flushing water would not cause significant changes to the water quality in the vicinity.

6.7.2.19 The proposed Reclaimed Water Service Reservoir for reuse of TSE as reclaimed water is a DP under Schedule 2 DPs requiring EPs. Hence, a separate EIA Study for proposed Reclaimed Water Service Reservoir will be conducted by the future project proponent.

Wastewater from Industrial and Commercial Activities

6.7.2.20 The wastewater generated from industrial and commercial activities such as operation of storage and workshop may have high concentration of pollutants which may potentially exceed the influent standards of the government sewage treatment plant. For individual commercial/industrial tenants, discharge license under WPCO will be required individually and the discharge standards according to TM-DSS to government foul sewers will be applied. In order to comply with the discharge standards to public sewerage, pre-treatment may be considered subject to the effluent characteristics. All the sewage and wastewater generated will then be conveyed to the proposed YLS STW for further treatment. No adverse water quality impact is anticipated.

6.7.3 Cumulative Impact with Concurrent Projects

6.7.3.1 The concurrent projects are discussed in **Section 1.8**. During the construction, a number of projects including Hung Shui Kiu New Development Area, Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station, and Improvement of Yuen Long Town Nullah would be located in the vicinity of the Project. The construction of EFTS (DP4), STW (DP7) and Reclaimed Water Service Reservoir (DP9) will be implemented by other studies. It is anticipated that general construction activities such as site formation and superstructure works would be involved, and potential impacts include site run-off, accidental chemical spillage, pollution from workforce sewage etc. may occur.

6.7.3.2 With the proper mitigation measures as listed in **Section 6.8.1** in place, including the adoption of ProPECC PN 1/94, provision of chemical portable toilets, etc., it is anticipated that the wastewater generated from the works sites would be managed in a proper manner without the cumulative effect with other projects.

6.7.3.3 However, for the works within watercourses, including construction box culvert and revitalisation of Yuen Long Nullah, it may have cumulative water quality impact due to site run-off and temporary reduction of the nullah capacity from the works of Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station, and Improvement of Yuen Long Town Nullah. According to the tentative construction programme stated in **Table 1.3**, the elevated pedestrian corridor will be completed in 2022, which will not overlap with the revitalisation of Yuen Long Nullah scheduled in Stage 2b (2023 – 2033) of the current proposed development. Also, Improvement of Yuen Long Town Nullah Stage 2 – Beautification Works would have construction works during 2025-2027. The Contractor should communicate with the responsible person of the Improvement of Yuen Long Town Nullah Stage 2 to avoid/ minimise the overlapping period as much as practicable. Nonetheless, with the implementation of proper precautionary measures recommended in **Section 6.8.1**, the potential water quality impact would be minimised and would not cause significant cumulative impact.

6.7.3.4 During operation phase, the proposed Reclaimed Water Service Reservoir for reuse of TSE as reclaimed water would be located at the southwest hillside of PDA Area 3. Although there is a potential service reservoir proposed under HSK EIA in close proximity, both reservoirs will adopt similar maintenance procedures as discussed in **Section 6.7.2.18** so that there would not be any discharge to the nearby water bodies. Hence, the cumulative water quality impact is not anticipated.

6.7.3.5 Also, the Project would not generate a net increase in pollution loading to the receiving water in the Deep Bay WCZ and the potential water quality impact of the sewage effluent in North Western WCZ would not be worse than that identified in SWSTW EIA as discussed in **Sections 6.7.2.4** to **6.7.2.10**. Hence, the cumulative impact with other concurrent projects is not anticipated.

6.8 Mitigation Measures

6.8.1 Construction Phase

General Site Operation

6.8.1.1 The following practice and measures should be implemented for all the construction works, including those for both DPs and non-DPs where applicable.

6.8.1.2 In accordance with the ProPECC PN 1/94, Environmental Protection Department, 1994, best management practices should be implemented as far as practicable as below:

- At the start of site establishment, perimeter cut-off drains to direct off-site water around the site should be constructed with internal drainage works. Channels (both temporary and permanent drainage pipes and culverts), earth bunds or sand bag barriers should be provided on site to direct stormwater to silt removal facilities.
- Diversion of natural stormwater should be provided as far as possible. The design of temporary on-site drainage should prevent runoff going through site surface, construction machinery and equipment in order to avoid or minimise polluted runoff. Sedimentation tanks with sufficient capacity, constructed from pre-formed individual cells of approximately 6 to 8 m^3 capacities, are recommended as a general mitigation measure which can be used for settling surface runoff prior to disposal. The system capacity shall

be flexible and able to handle multiple inputs from a variety of sources and suited to applications where the influent is pumped.

- The dikes or embankments for flood protection should be implemented around the boundaries of earthwork areas. Temporary ditches should be provided to facilitate the runoff discharge into an appropriate watercourse, through a silt/sediment trap. The silt/sediment traps should be incorporated in the permanent drainage channels to enhance deposition rates.
- The design of efficient silt removal facilities should be based on the guidelines in Appendix A1 of ProPECC PN 1/94. The detailed design of the sand/silt traps should be undertaken by the contractor prior to the commencement of construction.
- Construction works should be programmed to minimise surface excavation works during the rainy seasons (April to September). All exposed earth areas should be completed and vegetated as soon as possible after earthworks have been completed. If excavation of soil cannot be avoided during the rainy season, or at any time of year when rainstorms are likely, exposed slope surfaces should be covered by tarpaulin or other means.
- All drainage facilities and erosion and sediment control structures should be regularly inspected and maintained to ensure proper and efficient operation at all times and particularly following rainstorms. Deposited silt and grit should be removed regularly and disposed of by spreading evenly over stable, vegetated areas.
- If the excavation of trenches in wet periods is necessary, it should be dug and backfilled in short sections wherever practicable. Water pumped out from trenches or foundation excavations should be discharged into storm drains via silt removal facilities.
- All open stockpiles of construction materials (for example, aggregates, sand and fill material) should be covered with tarpaulin or similar fabric during rainstorms. Measures should be taken to prevent the washing away of construction materials, soil, silt or debris into any drainage system.
- Manholes (including newly constructed ones) should always be adequately covered and temporarily sealed so as to prevent silt, construction materials or debris being washed into the drainage system and storm runoff being directed into foul sewers.
- Precautions to be taken at any time of year when rainstorms are likely, actions to be taken when a rainstorm is imminent or forecasted, and actions to be taken during or after rainstorms are summarized in Appendix A2 of ProPECC PN 1/94. Particular attention should be paid to the control of silty surface runoff during storm events.
- All vehicles and plant should be cleaned before leaving a construction site to ensure no earth, mud, debris and the like is deposited by them on roads. An adequately designed and sited wheel washing facilities should be provided at every construction site exit where practicable. Wash-water should have sand and silt settled out and removed at least on a weekly basis to ensure the continued efficiency of the process. The section of access road leading to, and exiting from, the wheel-wash bay to the public road should be paved with sufficient backfall toward the wheel-wash bay to prevent vehicle tracking of soil and silty water to public roads and drains.

- Oil interceptors should be provided in the drainage system downstream of any oil/fuel pollution sources. The oil interceptors should be emptied and cleaned regularly to prevent the release of oil and grease into the storm water drainage system after accidental spillage. A bypass should be provided for the oil interceptors to prevent flushing during heavy rain.
- Construction solid waste, debris and rubbish on site should be collected, handled and disposed of properly to avoid water quality impacts.
- All fuel tanks and storage areas should be provided with locks and sited on sealed areas, within bunds of a capacity equal to 110% of the storage capacity of the largest tank to prevent spilled fuel oils from reaching water sensitive receivers nearby.
- Regular environmental audit on the construction site should be carried out in order to prevent any malpractices. Notices should be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the water bodies, marsh and ponds.

6.8.1.3 By adopting the best management practices, it is anticipated that the impacts of general site operation will be reduced to satisfactory levels before discharges. The details of best management practices will be highly dependent to actual site condition and the Contractor shall apply for a discharge license under WPCO.

Prevention of Accidental Spillage of Chemicals

6.8.1.4 The chemicals used during construction, such as fuel, oil, solvents and lubricants shall be properly stored and contained in designated area with secondary containment to prevent spillage and contamination of the nearby water environment.

6.8.1.5 Any maintenance activities and workshops with chemicals use shall be located away from watercourses on hard standings within a bunded area. Sumps and oil interceptors should be provided as appropriate.

6.8.1.6 The Contractor shall register as a chemical waste producer and employ licensed collector for collection of chemical waste from the construction site. Any chemical waste generated shall be managed in accordance with the Waste Disposal (Chemical Waste) (General) Regulation.

Sewage from Workforce

6.8.1.7 Portable chemical toilets and sewage holding tanks should be provided for handling the construction sewage generated by the workforce. A licensed contractor should be employed to provide appropriate and adequate portable toilets to cater 0.23 m^3/day/employed population and be responsible for appropriate disposal and maintenance.

6.8.1.8 Notices should be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the nearby environment during the construction phase of the Project. Regular environmental audit on the construction site should be conducted in order to provide an effective control of any malpractices and achieve continual improvement of environmental performance on site. It is anticipated that sewage generation during the construction phase of the Project would not cause water quality impact after undertaking all required measures.

Contaminated Groundwater and Site Runoff

6.8.1.9 To prevent the water quality due to the contaminated water from the area with contamination, the following mitigation measures should be adopted.

- Cover the contaminated soil and surface to prevent the generation of contaminated water.
- No open stockpiling of contaminated soil should be allowed to prevent generation of contaminated water due to precipitation.
- Contaminated water, either from groundwater or runoff, should be treated by wastewater treatment facility (WTF) to an acceptable level as indicated in TM-DSS before disposal if the deployment of such WTF is feasible.
- Recharging the contaminated groundwater back to the aquifer should be sought if treatment of the contaminated groundwater by WTF is not feasible, subject to the agreement with EPD.

Construction Works near/ within Watercourses

6.8.1.10 Apart from the general site best management practices, extra care shall be paid for works near watercourses to minimise the potential water quality impacts, especially for the construction of deck over Yuen Long Nullah and revitalisation works of Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah. The measures described in ETWB TC (Works) No. 5/2005 "Protection of natural streams / rivers from adverse impacts arising from construction works" should be adopted where applicable. The major measures are list below:

- Cofferdams and impermeable sheet piles should be installed as appropriate to isolate the flow of the nullah from the construction works area. The detailed design of the cofferdams will be conducted by the Contractor during the construction phase to fulfil the requirements in DSD Technical Circular No. 1/2017 "Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater Drainage System" for DSD approval, in order to formulate feasible options of these temporary structure.
- Stockpiling of construction materials and dusty materials should be located away from any watercourses, contained in bunded areas and covered with tarpaulin.
- Construction debris and spoil should be covered with tarpaulin during storage. Timely removal of materials away from the site for disposal should be arranged to avoid being washed into the nearby watercourses.
- Water pumps should be used to collect any wastewater and construction site surface runoff within the cofferdam/ temporary working platform. The collected wastewater shall be properly treated before discharge.
- Toe-board and bunds shall be provided along the edge of the works area/ temporary platform to prevent wastewater/ debris from falling into the watercourses.
- Any temporary works site inside the watercourses should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the water quality.
- Proper shoring may need to be erected in order to prevent soil / mud from slipping into the inland water bodies.

- Construction effluent, site run-off and sewage should be properly collected and/or treated.

Removal/ Diversion of watercourses

6.8.1.11 During removal and diversion of watercourse, precaution measures shall be implemented to prevent adverse water quality impact to the surrounding environment. Good site practices as described in ETWB TC (Works) No. 5/2005 "Protection of natural streams / rivers from adverse impacts arising from construction works" and ProPECC PN 1/94 "Construction Site Drainage" should be adopted where applicable. The following major measures shall be implemented:

- Cofferdams and impermeable sheet piles should be installed as appropriate to isolate the water flow from the construction works area.
- Dewatering or flow diversion shall be conducted prior to the construction works to prevent water overflow to the surrounding area.
- Watercourse removal and flow diversion should be conducted in dry season as far as practicable when the water flow is low.
- Water drained from the watercourse shall be diverted to new/ temporary drainage for watercourse diversion. For watercourse removal, the water drained shall be collected and treated to meet the requirements of WPCO and TM-DSS before discharge.
- Any excavated land-based sediment from the removal/ diversion of watercourse shall be properly stored at bunded areas away from any watercourse and covered with tarpaulin before transporting out of the site. Detailed management of excavated land-based sediment is discussed in **Section 11.5.1**.

Removal/ Filing of ponds

6.8.1.12 During ponds filling and removal, the following precaution measures shall be implemented to prevent adverse water quality impact to the surrounding environment.

- Dewatering shall be conducted prior to the construction works to prevent water overflow to the surrounding area.
- Water drained from the ponds shall be collected in appropriate temporary storage tank and reuse on-site as far as practicable. Surplus drained water shall be properly disposal at STW. No direct discharge to stormwater drainage system or marine water should be allow.
- Any excavated land-based sediment from the ponds shall be properly stored at bunded areas away from any watercourse and covered with tarpaulin before transporting out of the site. Detailed management of excavated land-based sediment is discussed in **Section 11.5.1**.

6.8.2 Operational Phase

Emergency Discharge from Sewage Treatment Works and Sewage Pumping Stations

6.8.2.1 As recommended in **Section 7.6.5**, given the sensitivity of inner Deep Bay in term of water quality and ecology, extensive effort will be expedited to avoid the occurrence for emergency discharge. In order to achieve this, the design of SPSs and

STW will be cautiously reviewed to include additional provisions including as follows:

- Twin rising mains would be provided. Should one of the duty mains be taken out of operation, the remaining one would still be able to deliver flow;
- Standby pumps and treatment facilities would be provided in case of unexpected breakdown of pumping and treatment facilities such that the standby pumps and treatment facilities could take over and function to replace the broken pumps; and
- Dual electricity supply or backup power supply facilities such as diesel generator would be provided in case of power failure to sustain the function of pumping and treatment facilities.

6.8.2.2 With the implemented mitigation measures, significant impact due to emergency sewage overflow from the proposed STW and SPSs is not anticipated.

Change in Drainage System and Road Runoff

6.8.2.3 During the operational phase, vehicle dust, tyre scraps and oils might be washed away from the road surface / open areas to the nearby water courses by surface runoff or road surface cleaning. Subject to detailed design and requirement of relevant government departments, the capacities of road drainage system shall cater the runoff from 50 year-return-period rainstorm. Proper drainage systems with silt traps and oil interceptors should be installed. The design of road gullies with silt traps should be incorporated in later detailed design.

6.8.2.4 Runoff will be controlled by best management practice. Runoff will be intercepted by properly designed and managed silt traps at appropriate spacing so that common roadside debris, refuse and fallen leaves etc. can be captured before allowing the runoff to drain into watercourses such as Yuen Long Creek. At the outlets to watercourses, the project proponent or the delegated operation parties should manage the road/open area cleaning prior to the occurrence of a storm. Moreover, it is recommended each of the cleaning events should be carried out during low traffic flow period, preferably using either manual methods or mechanical means such as vacuum sweeper/truck equipped with side broom, which is to sweep road sludge and debris into the suction nozzle to increase the removal efficiency of pollutants. The collected pollutants would be tankered away for off-site disposal at landfill sites. After the removal of the pollutants, the pollution levels from stormwater would be much reduced.

6.8.2.5 Given the intermittent nature of non-point source pollution and adopting flexible management to suit site conditions, the impact to the receiving water body is insignificant.

Maintenance Flushing for Reclaimed Water Service Reservoir

6.8.2.6 In order to avoid the water quality impact caused by the maintenance flushing, all the effluent generated during the process would be avoided to discharge into the catchment by implementation of the following measures.

- Before the flushing, the reclaimed water inside the reservoir would be discharged to the foul sewers and the sludge remained at the bottom will be collected and disposed by a licensed waste collector.

- During the flushing, the cleansing water would be collected by a licensed waste collector via the washout pipes at a controlled manner to avoid the possible impact to the nearby streams with ecological importance.

Wastewater from Storage and Workshop Area

6.8.2.7 For individual commercial/industrial tenants, discharge license under WPCO will be required individually and the discharge standards according to TM-DSS to government foul sewers will be applied. In order to comply with the discharge standards to public sewerage, pre-treatment may be considered subject to the effluent characteristics. Compliance with WPCO for discharge of wastewater will thus be ensured.

6.8.2.8 Also, in order to prevent potential environment impact due to accidental spillage/ discharge of untreated wastewater, stop-logs should be installed where appropriate in the drainage system. Hence, the drainage system within the storage and workshop area can be isolated from the general drainage system during accidental events.

6.9 Enhancement Measures

6.9.1 Operational Phase

Livestock Waste generated from Pig Farms and Chicken Farms

6.9.1.1 As discussed in **Section 1.7**, there are currently 3 pig farms and 3 chicken farms in the PDA whose effluent would be discharged into nearby watercourses. Release of livestock waste into nearby watercourse would increase the nutrient level in the water environment, deteriorating the water quality of these watercourses. However, as 3 pig farms and 2 chicken farms will be decommissioned, the pollution loading into these watercourse and the Deep Bay WCZ would be significantly reduced after the proposed development. This would be recognized as the pollution credit gained from the Project.

Unsewered Population

6.9.1.2 As discussed in **Section 1.7**, there is no sewerage within the PDA, and the watercourses are subject to discharges from the unsewered population. With the proposed sewage treatment plant implemented under the Project, all the areas within the PDA boundary will be provided with sewers that convey the sewage to the new sewage treatment plant. Hence, all of the unsewered population within the PDA will be sewered and the pollution loading from the unsewered population will be abated. This would result in a pollution credit gained from the proposed development.

Conservation of Watercourses with Ecological Importance

6.9.1.3 There are three watercourses with ecological importance (i.e. WSRs S02, S03 and S04 as shown in **Figure 6.1**) zoned as ‘GB’ for conservation purpose and buffer zone. This zone mainly covers the three land corridors adjoining the watercourses with ecological importance within the PDA. As discussed in the ecology chapter (**Chapter 8**), it is intended to preserve the integrity of these watercourses and their riparian corridors which are largely natural/semi-natural with good water quality, contain several species of conservation concern and have high ecological value providing linkages and corridors. These watercourses have good species diversity and assemblages and they act as nurseries for several aquatic species. Pollution to these watercourses shall be avoided as discussed in **Section 8.7.4**.

Further Polishing of TSE by Reedbed

6.9.1.4 In the previous project namely Planning and Engineering Study on Development of Lok Ma Chau Loop and existing Tin Shui Wai Wetland Park, it has been proven that reedbed is capable to further polish the TSE by means of biological breakdown process. There are a total of approximate 4 ha area adjacent to YLS STW in the RODP as shown in **Figure 7.1**, which would be developed into a reedbed. A small amount of TSE from YLS STW will pass through the reedbed for further polishing before discharging into Yuen Long Nullah. More details can be found in **Section 7.6.3** and **Appendix 7.1**

6.10 Residual Impacts

6.10.1.1 With implementation of the recommended mitigation measures for drainage and sewerage system, the proposed development would not substantially generate unacceptable residual water quality impacts.

6.11 Conclusions and Recommendations

6.11.1.1 During the construction phase, with full implementation of the mitigation measures, no substantial impact is anticipated from surface runoff from construction site and sewage generated from construction workforce in construction phase.

6.11.1.2 During the operational phase, a new STW will be proposed and designed to achieve a total treatment capacity of 24,000m^3/day in ADWF and up to tertiary level treatment standard to enable reuse of TSE as reclaimed water while the rest of sewage is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to the Urmston Road submarine outfall. Among the reclaimed water, most of it will be reused for non-potable uses such as toilet flushing in the PDA and in adjacent developments; and only a small amount of TSE will be discharged to a reedbed adjoining to the proposed YLS STW for further polishing before entering Yuen Long Nullah. As discussed in **Sections 6.7.2.4** to **6.7.2.9,** the Project would not generate a net increase in pollution loading to the receiving water in the Deep Bay WCZ. Hence, the policy "No Net Increase in Pollution Loads Requirement in Deep Bay" would be complied with. Also, the Project would also bring benefits by providing sewerage infrastructure to the existing unsewered areas. Hence, the policy "No Net Increase in Pollution Loads Requirement in Deep Bay" would be complied with. Also, the Project would also bring benefits by providing sewerage infrastructure to the existing unsewered areas within the PDA.

6.11.1.3 Contingency measures such as twin rising mains, standby pump and treatment facilities and dual electricity supply or backup power supply facilities as stated in **Section 6.8.2.1** would be adopted for the proposed YLS STW and two SPSs to prevent emergency situation as far as practicable. In the very unlikely case that all these measures fail, the discharge from the proposed YLS STW and two SPSs will be delivered to the nearby proposed reedbed and the nearby channelised Yuen Long Nullah respectively. However, the water quality impact due to emergency discharge is anticipated to be short-term and thus insignificant. For other potential water pollution sources arising from the Project such as maintenance flushing of Reclaimed Water Service Reservoir, wastewater from industrial and commercial activities, etc., no significant water quality impact is anticipated with the implementation of mitigation measures in **Section 6.8.2**.

6.11.1.4 The project proponent would further study the feasibility of reusing the reclaimed water from the Reclaimed Water Reservoir for toilet flushing in Tin Shui Wai Area.

6.11.1.5 No substantial residual impact is anticipated during both construction and operational phases of the Project.

6.12 References

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